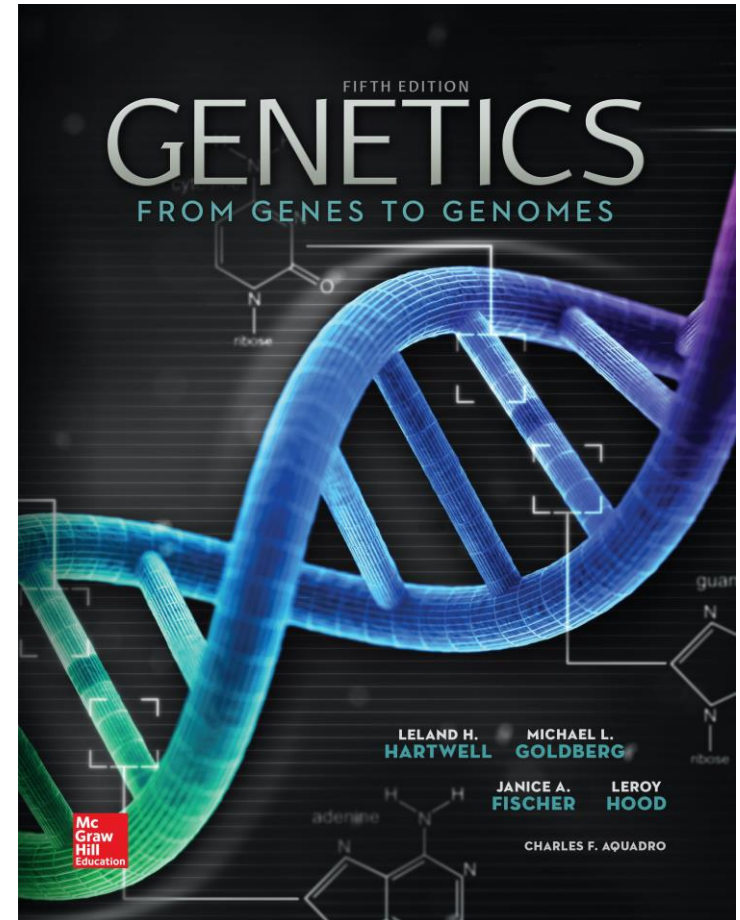


Genetics: From Genes to Genomes

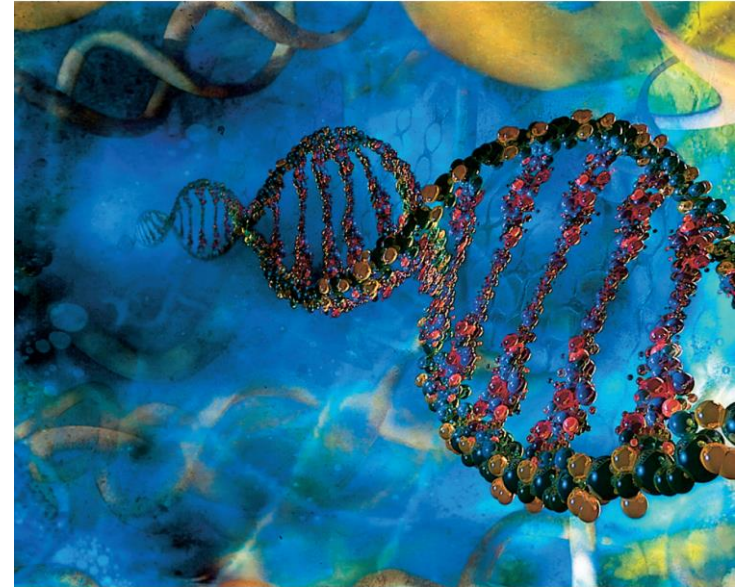
Fifth Edition

Leland H. Hartwell, Michael L. Goldberg, Janice A. Fisher, Leroy Hood, and Charles F. Aquadro

**Prepared by Kristin D. Patterson
University of Texas at Austin**



DNA Structure, Replication, and Recombination



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CHAPTER OUTLINE

- 6.1 Experimental Evidence for DNA as the Genetic Material
- 6.2 The Watson and Crick Double Helix Model of DNA
- 6.3 Genetic Information in DNA Base Sequence
- 6.4 DNA Replication
- 6.5 Recombination at the DNA Level

Two general themes to genes at the molecular levels

The genetic functions of DNA flow directly from its molecular structure

- **Knowledge of molecular structure of DNA makes it possible to understand biochemical processes of genetics**

All of the genetics functions of DNA depend on specialized proteins that "read" the information in DNA sequence

- **DNA itself is chemically inert**

6.1 Experimental evidence for DNA as the genetic material

Learning objectives

- **Describe the chemical components of DNA**
- **Summarize the methods by which DNA was found to be located in chromosomes.**
- **Explain the bacterial transformation experiments of Avery and his colleagues and the results they obtained.**
- **Describe the “blender” experiments of Hershey and Chase and what they revealed about DNA versus protein.**

Chemical studies located DNA in the chromosomes

F. Meischer (1869) extracted "nuclein" from nuclei of human white blood cells

- **Weakly acidic, phosphorus rich material**

Chemical analysis of nuclein revealed that its major component was deoxyribonucleic acid (DNA)

- **Contains deoxyribose, found in nucleus, and is acidic**

Staining of cells revealed that DNA localized almost exclusively within chromosomes

- **Schiff reagent – stains DNA red**

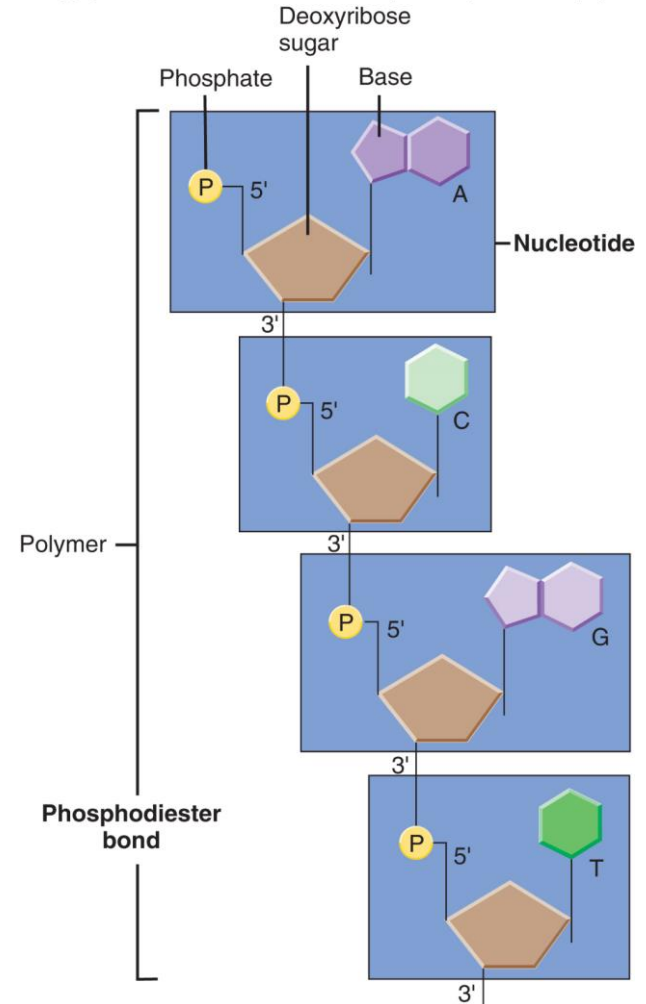
The chemical composition of DNA

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DNA contains four kinds of nucleotides linked in a long chain

Phosphodiester bonds – covalent bonds joining adjacent nucleotides

Polymer – linked chain of subunits



Are genes composed of DNA or protein?

DNA is made of only four different subunits

- **Too simple to specify genetic complexity?**

Protein is made of 20 different subunits

- **More potential for creating different combinations?**
- **Chromosomes contain more protein than DNA**

Bacterial transformation implicates DNA as the substance of genes

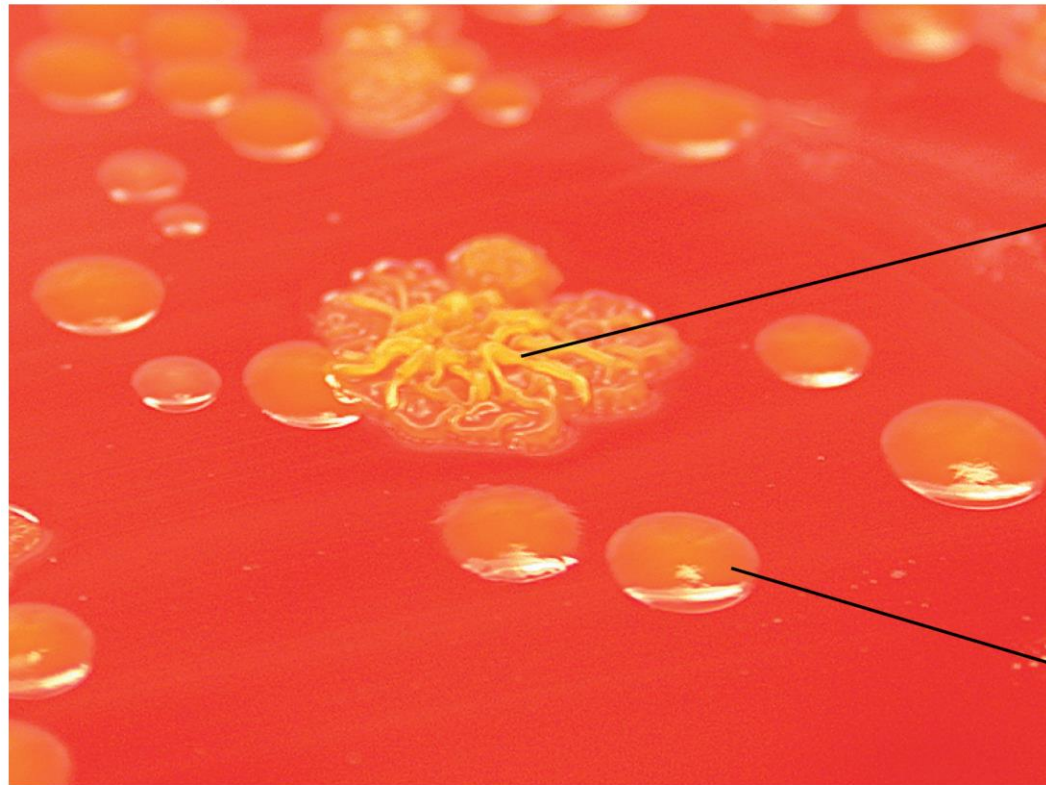
F. Griffith (1928) did experiments with two strains of *Streptococcus pneumoniae*

- **Differ in colony morphology and biological activity**
- **Smooth (S) strain – virulent**
- **Rough (R) strain – nonvirulent**
- **R cells could be transformed by genetic material transferred from dead S cells**

Avery, MacLeod, and McCarty (1944) provided evidence that DNA is the "transforming principle" of S cells

Two forms of *S. pneumoniae* have different colony morphologies

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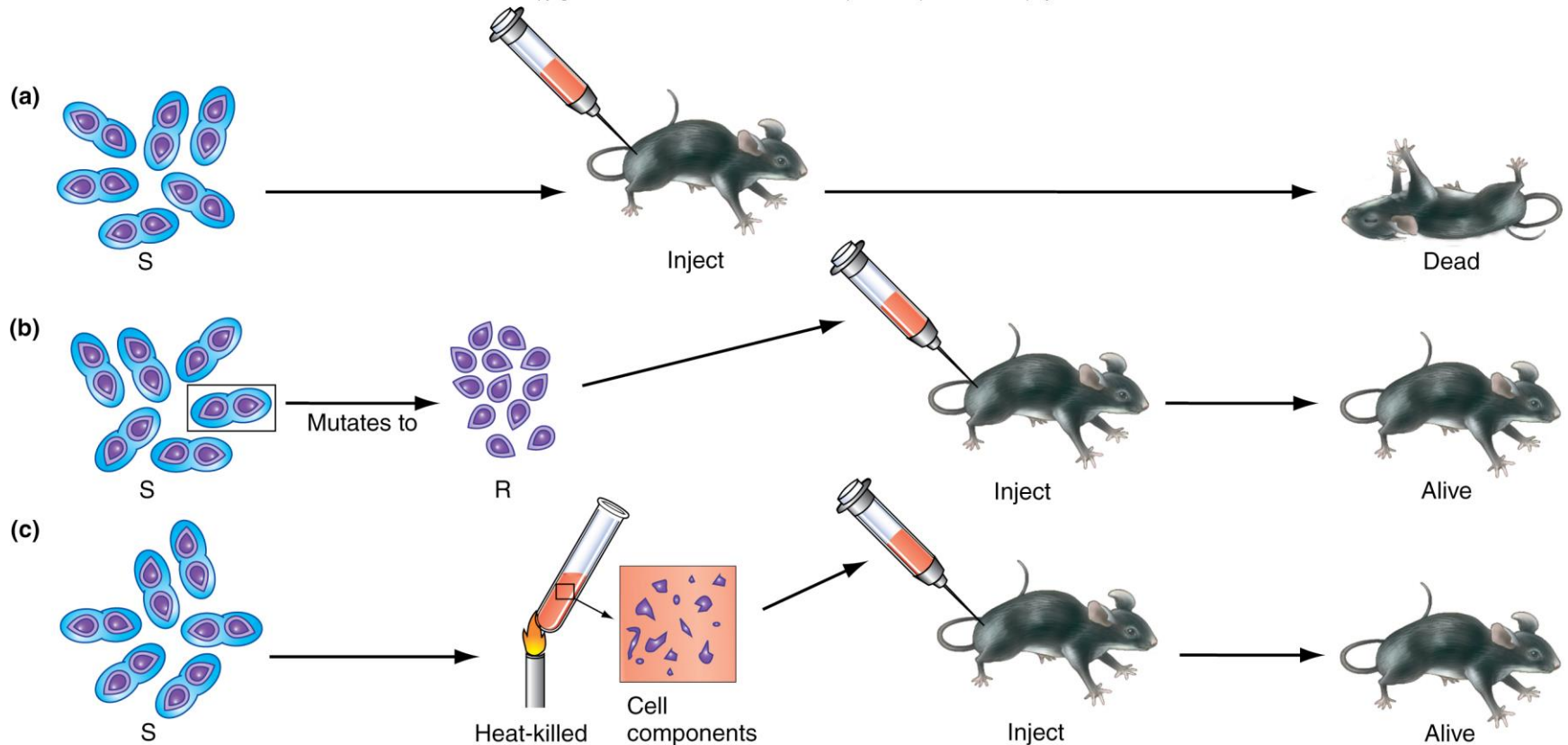
Rough colony

Smooth colony

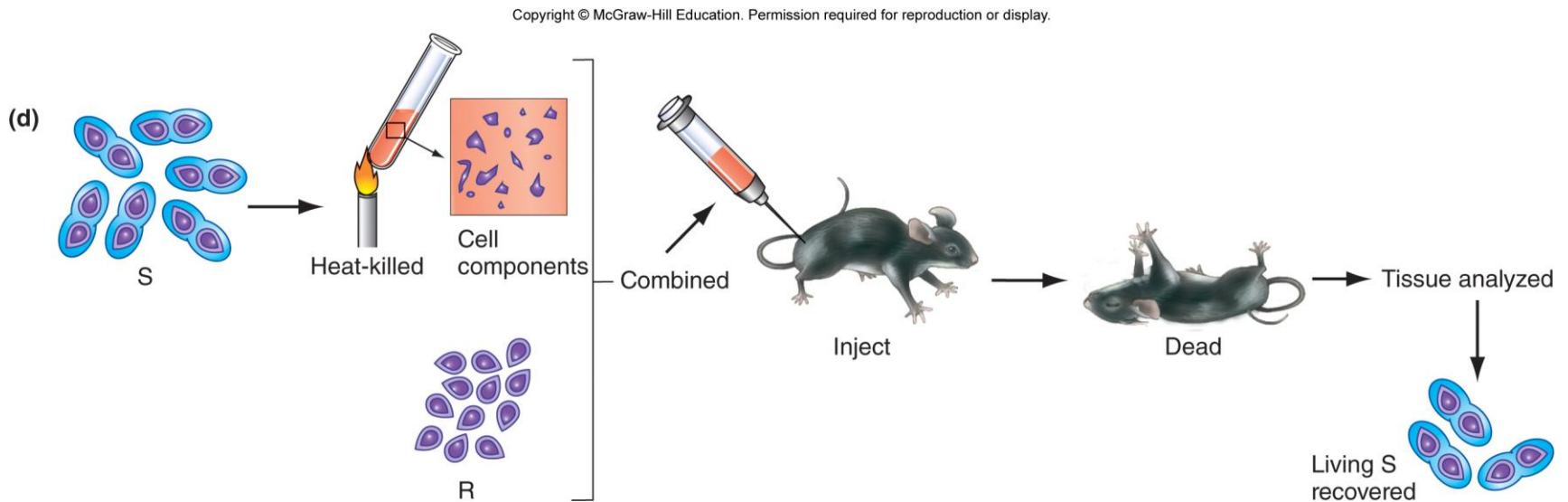
Used with permission from: Arnold et al; New Associations With Pseudomonas Luteola Bacteremia: A Veteran With A History Of Tick Bites And A Trauma Patient With Pneumonia. The Internet Journal of Infectious Diseases 2005 : Volume 4 Number 2. © & Courtesy of Dr. Forest Arnold, University of Louisville;

The two forms of *S. pneumoniae* differ in their effects on mice

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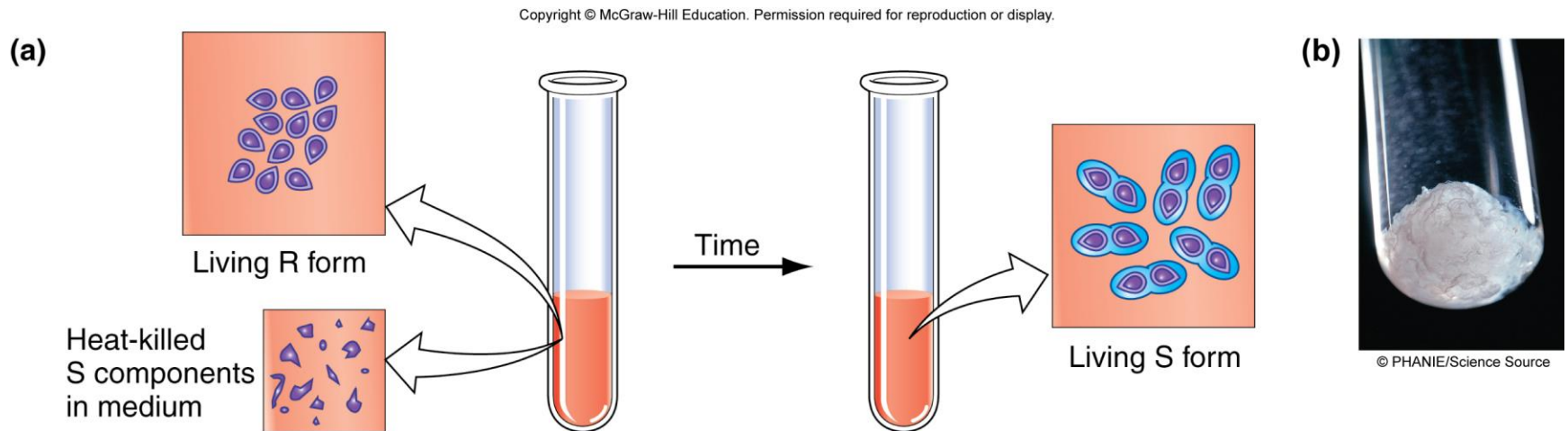


Griffith's experiment that provided evidence of transformation



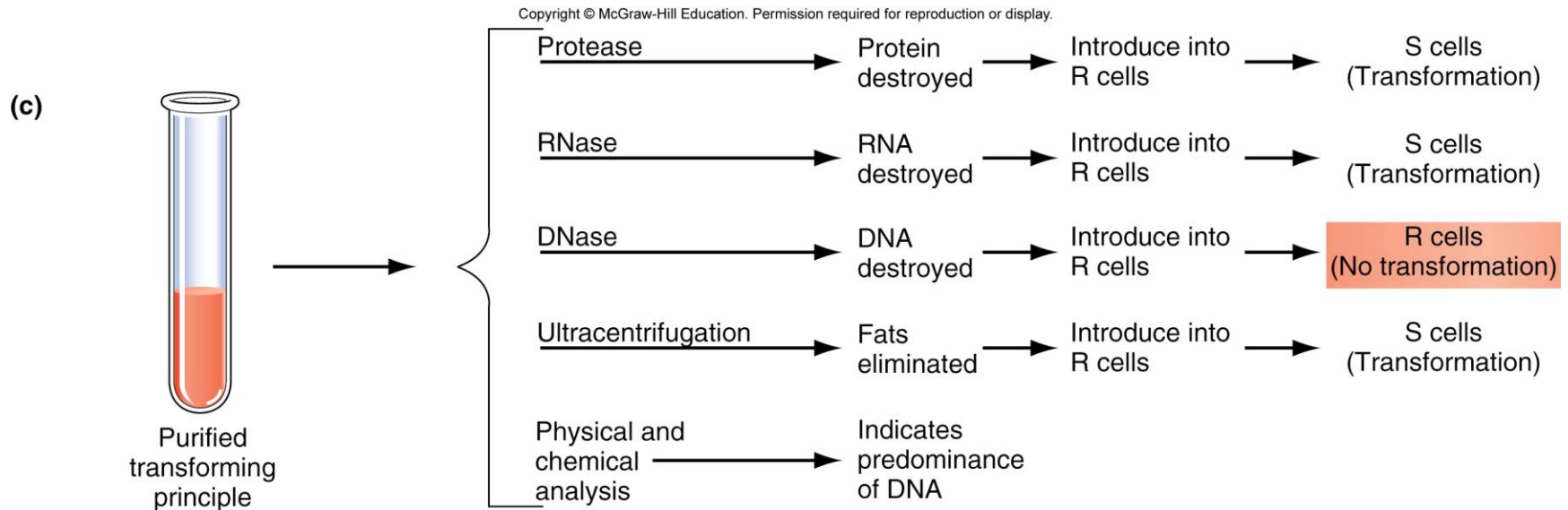
Something from heat-killed S cells
transformed R cells to S cells

Avery, MacLeod, and McCarty identified the transforming principle



Medium from heat-killed S cell culture transforms R cells.
The transforming principle was purified from extracts.

Avery, MacLeod, and McCarty confirmed that DNA is the transforming principle

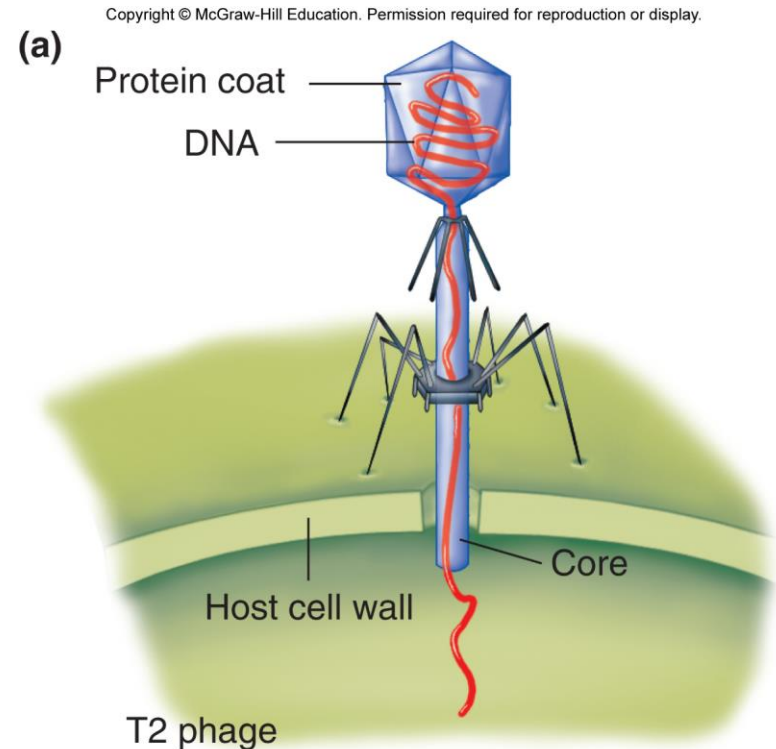


Not everyone was convinced by these data.

A simple system to test whether protein or DNA is the genetic material

Bacteriophages (phages) are viruses that infect bacteria

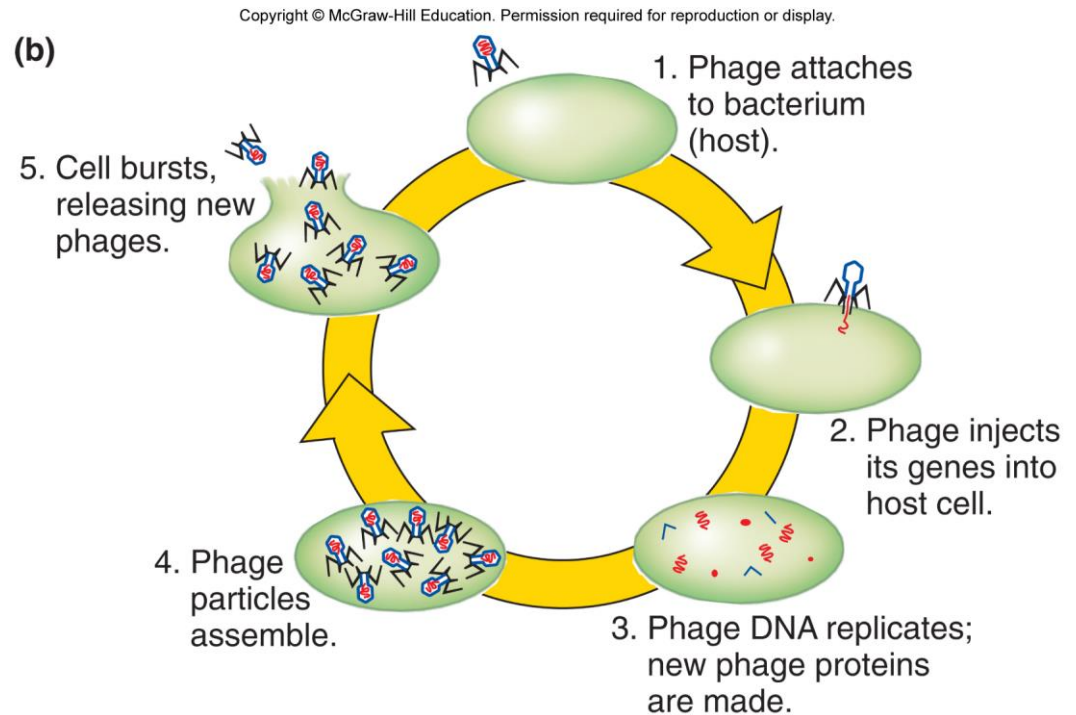
Phage particles contain roughly equal amounts of protein and DNA



The life cycle of bacteriophage T2

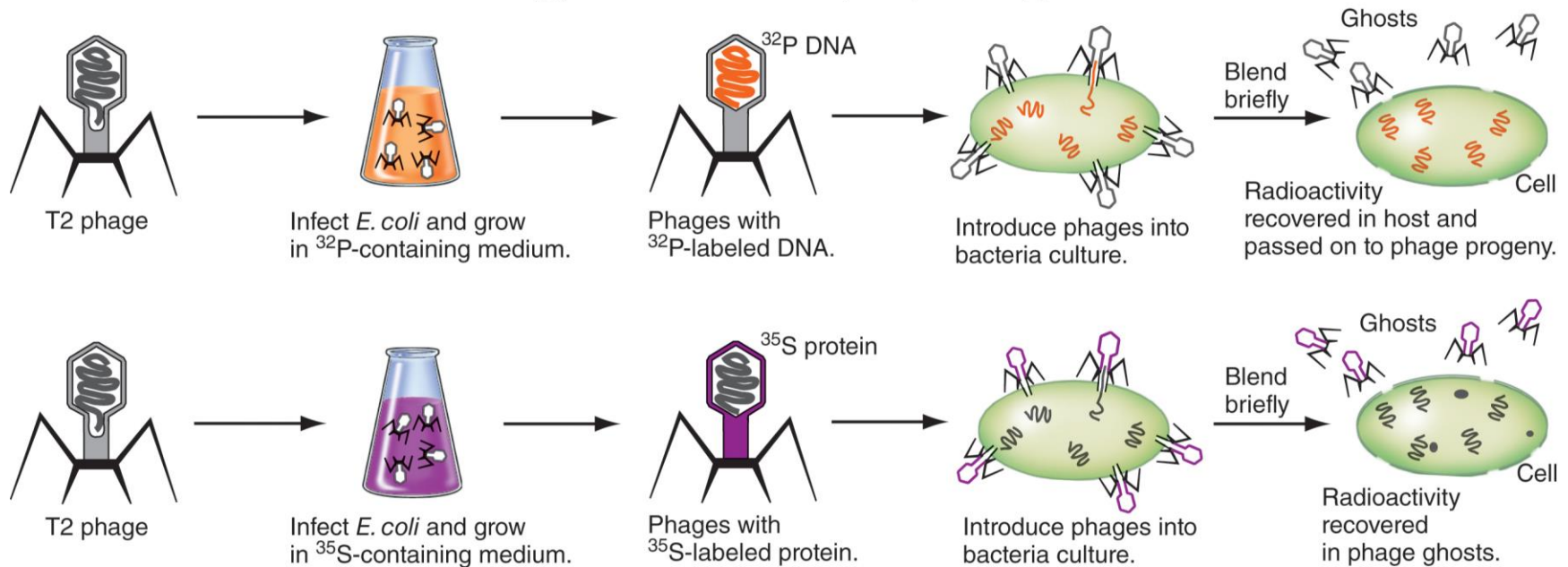
Contain very few genes but able to replicate themselves inside bacterial host

- **After infection, "ghost" of phage particle remains attached to outer surface of cell**
- **Phage genetic material "injected" into bacterial cell**



The Hershey-Chase Waring blender experiment

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6.2 The Watson and Crick double helix model of DNA

Learning objectives:

- Describe the key features of the Watson-Crick model for DNA structure.
- Explain what is meant by the antiparallel polarity of the two DNA strands.
- Distinguish the different structural forms of DNA from one another.

The basis of the Watson-Crick double helix model of DNA

R. Franklin and M. Wilkins (1952) solved X-ray diffraction pattern of DNA (see Figure 6.8)

- **DNA is helical structure with 20 Å diameter**
- **Spacing between repeating units is 3.4 Å**
- **Helix undergoes a complete turn every 34 Å**

Detailed knowledge of chemical constituents of DNA

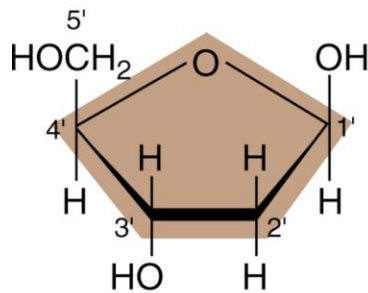
- **Four nitrogenous bases [guanine (G), adenine (A), cytosine (C), and thymine (T)], deoxyribose, phosphate**

E. Chargaff – base composition of DNA from many organisms

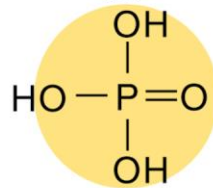
- **Ratios of bases: A:T ratio is 1:1, G:C ratio is 1:1**

A detailed look at DNA's chemical constituents

Deoxyribose



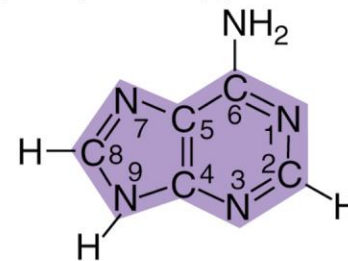
Phosphate



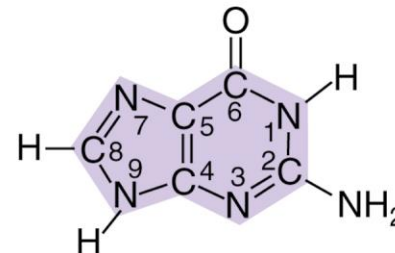
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Four nitrogenous bases

Purines

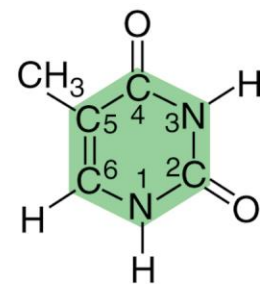


Adenine (**A**)

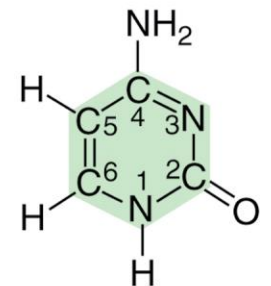


Guanine (**G**)

Pyrimidine



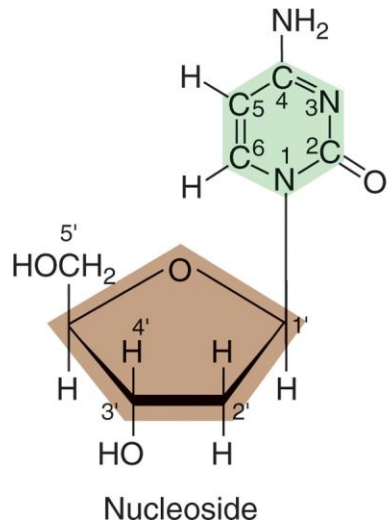
Thymine (**T**)



Cytosine (**C**)

A detailed look at DNA's chemical constituents (cont)

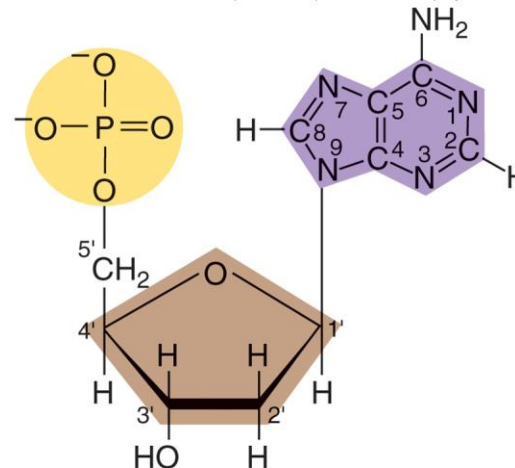
Attachment of base to sugar



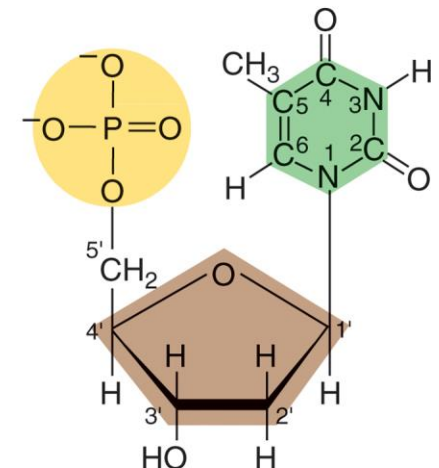
Addition of phosphate to nucleoside

Purine nucleotide

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Pyrimidine nucleotide



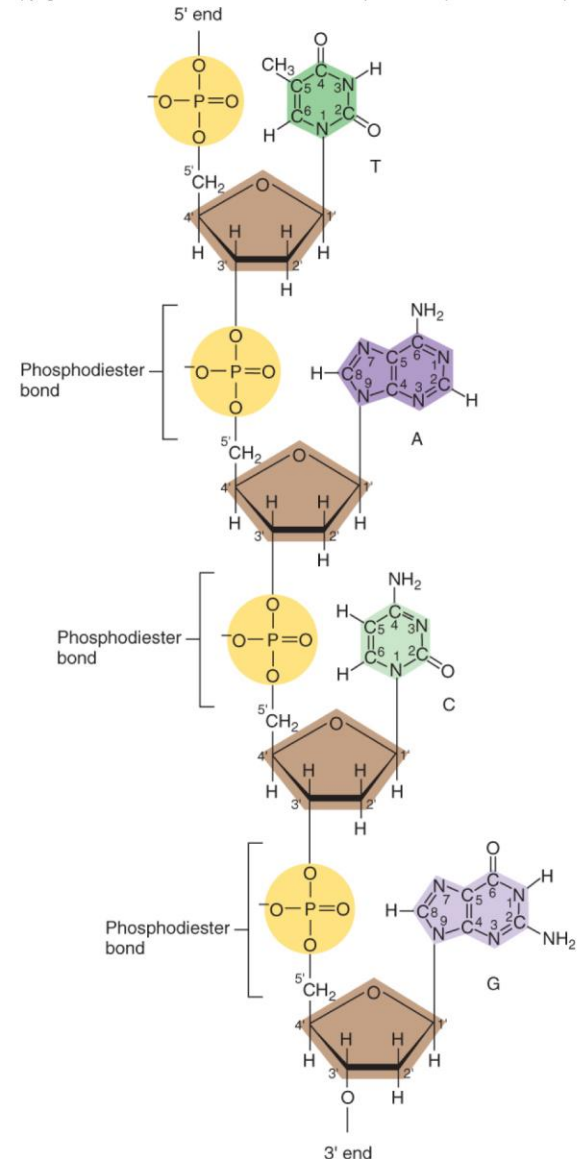
A detailed look at DNA's chemical constituents (cont)

Nucleotides linked in a directional chain

Phosphodiester bonds always form covalent link between 3' carbon of one nucleotide and 5' carbon of the next nucleotide

Note the 5'-to-3' polarity

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Chargaff's data on nucleotide base composition in the DNA of various organisms

In all organisms, ratios of A to T and G to C are roughly 1:1

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TABLE 6.1

Chargaff's Data on Nucleotide Base Composition in the DNA of Various Organisms

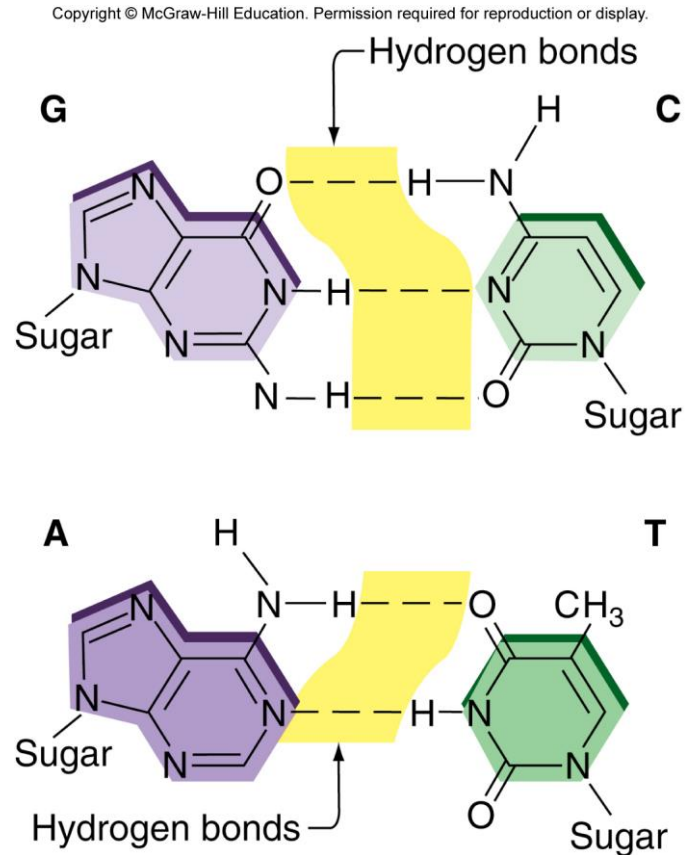
Organism	Percentage of Base in DNA				Ratios	
	A	T	G	C	A:T	G:C
<i>Escherichia coli</i>	24.7	23.6	26.0	25.7	1.05	1.01
<i>Saccharomyces cerevisiae</i>	31.3	32.9	18.7	17.1	0.95	1.09
<i>Caenorhabditis elegans</i>	31.2	29.1	19.3	20.5	1.07	0.96
<i>Drosophila melanogaster</i>	27.3	27.6	22.5	22.5	0.99	1.00
<i>Mus musculus</i>	29.2	29.4	21.7	19.7	0.99	1.10
<i>Homo sapiens</i>	29.3	30.0	20.7	20.0	0.98	1.04

Complementary base pairing

Base pairs consist of hydrogen bonds (weak electrostatic bonds) between a purine and a pyrimidine (G with C, A with T)

Consistent with Chargaff's rules

Each base pair has ~ same shape



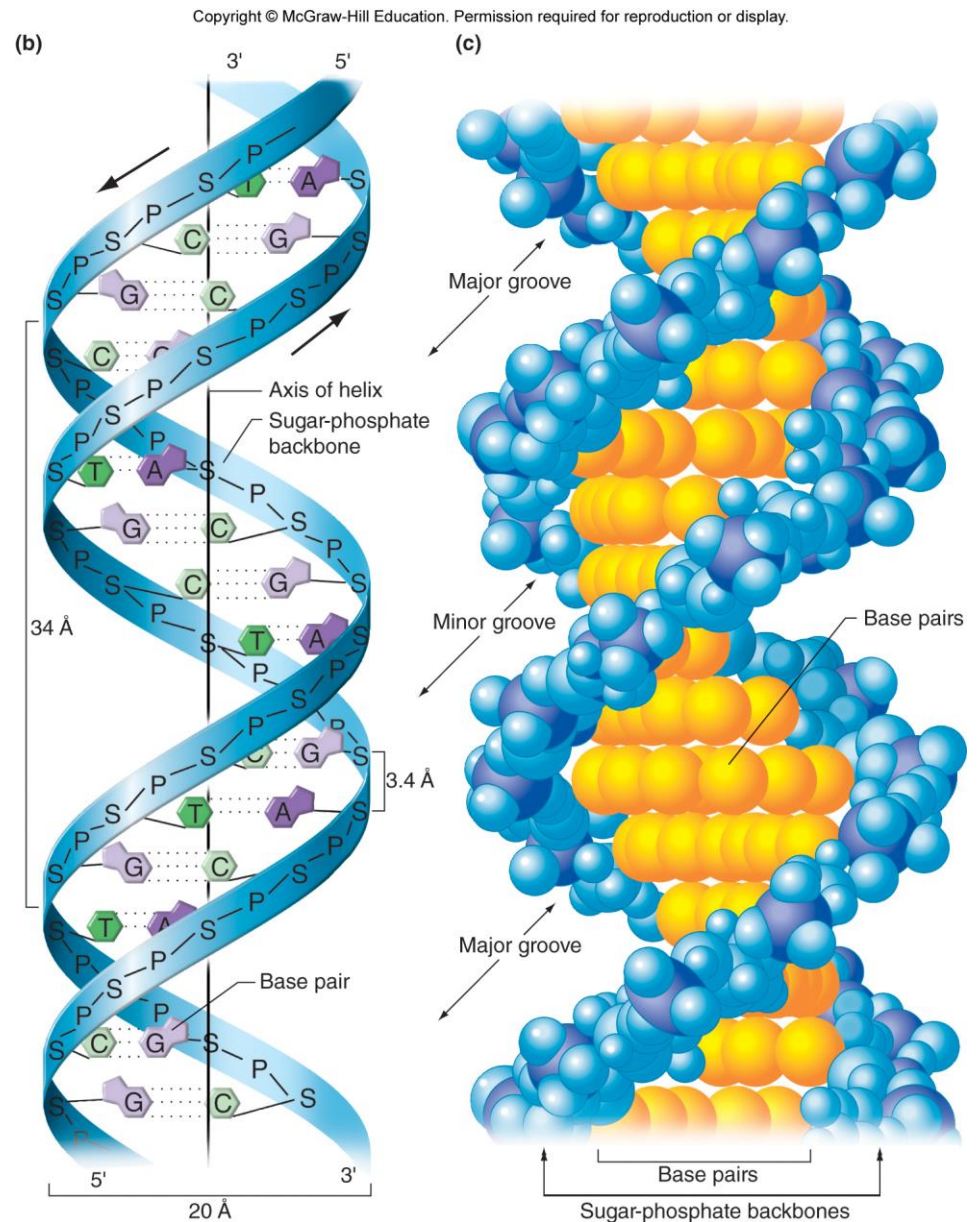
The double helix structure of DNA

Strands are antiparallel

Sugar – phosphate backbone on the outside

Base pairs in the middle

Two chains held together by H bonds between A-T and G-C base pairs

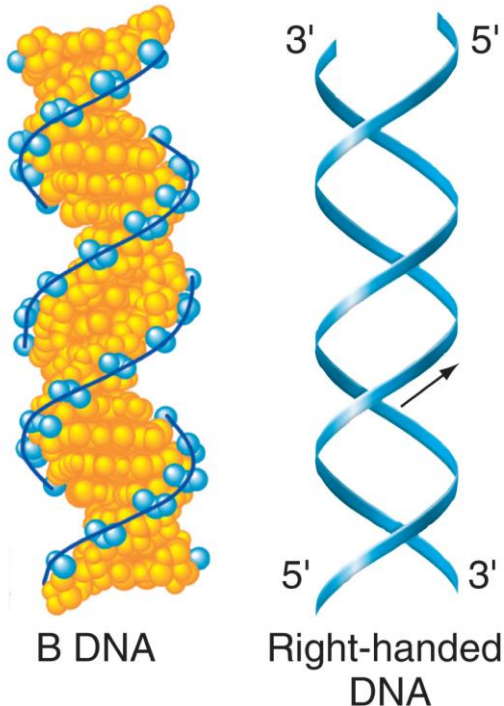


Z DNA is one variant of the double helix

B-form DNA forms right-handed helix and has a smooth backbone

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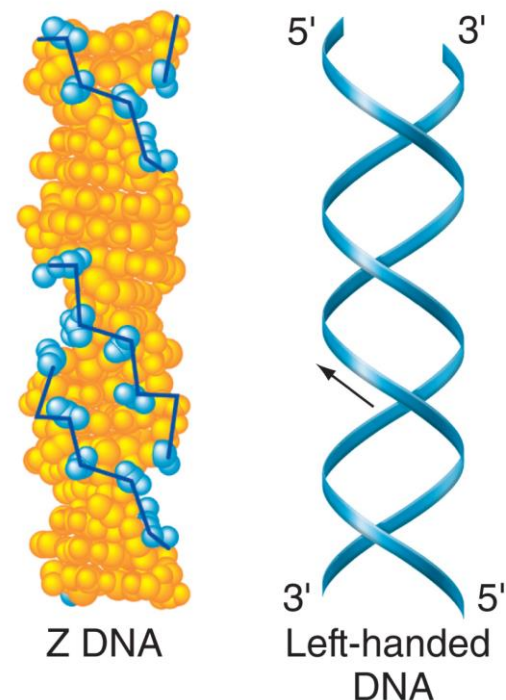
(a)



Z-form DNA forms left-handed helix and has an irregular backbone

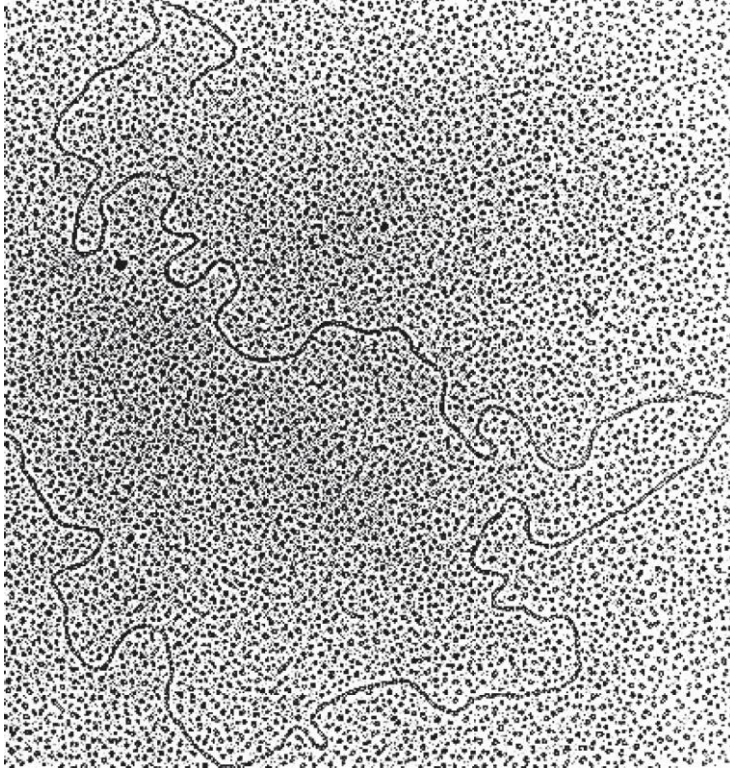
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(b)



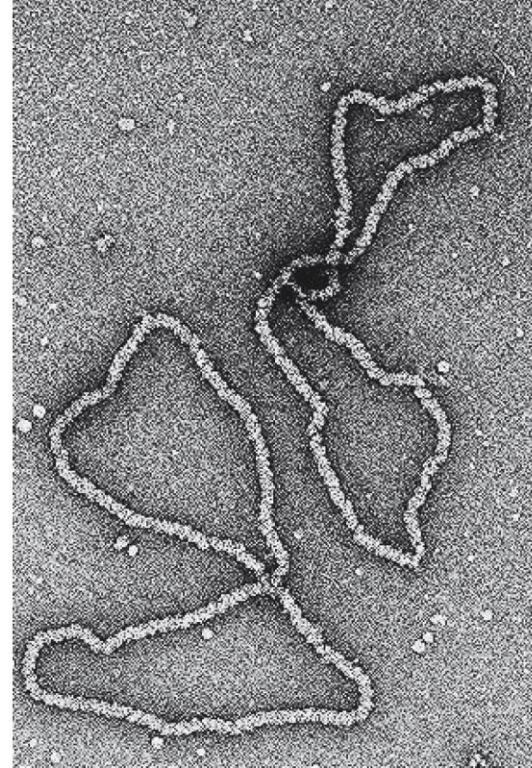
DNA can be linear or circular, double- or single-stranded

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**Linear double-stranded
human chromosome**

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**Circular single-stranded
phage M13 chromosome**

Four questions about how DNA structure relates to genetic functions

How does the molecule carry information?

- **Base sequence**

How is that information is copied for transmission to future generations?

- **DNA replication**

What mechanisms allow the genetic information to change?

- **Recombination**
- **Mutations (chapter 7)**

How does DNA-encoded information govern the expression of phenotype?

- **Gene functions (chapter 8)**

6.3 Genetic information in nucleotide sequence

Learning objectives:

- Explain how DNA stores complex information
- Compare the two ways in which the information in DNA may be accessed by proteins.
- Describe the structural differences between DNA and RNA.

DNA stores information in the sequence of its bases

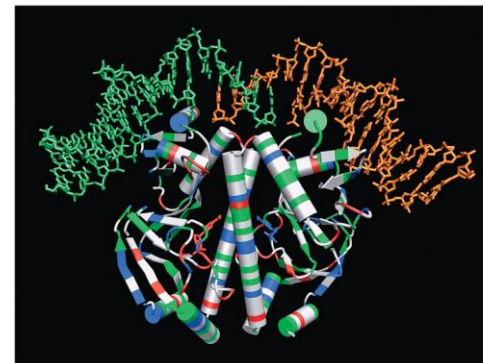
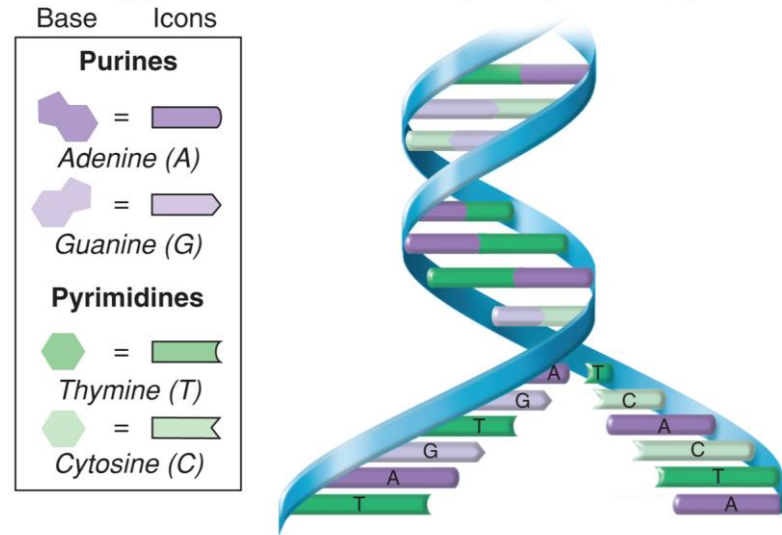
Most genetic information is
"read" from unwound DNA

- synthesis of DNA or RNA

Some genetic information is
accessible within double-
stranded DNA

- DNA-binding proteins that
regulate gene expression

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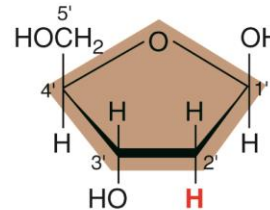
Chemical constituents of RNA

Three major chemical differences between RNA and DNA

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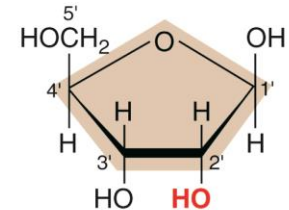
DNA

- Deoxyribose

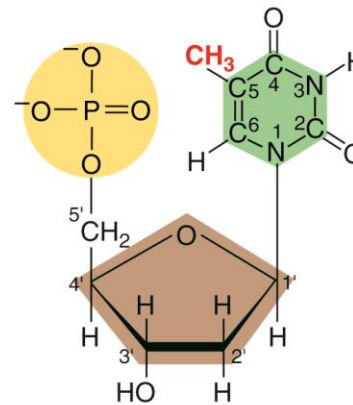


RNA

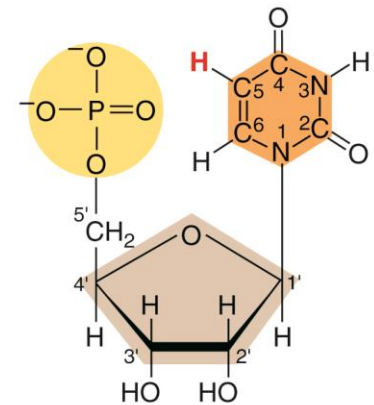
- Ribose



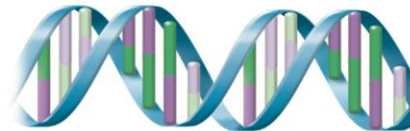
- Thymine deoxyribonucleotide



- Uracil ribonucleotide



- Usually double-stranded



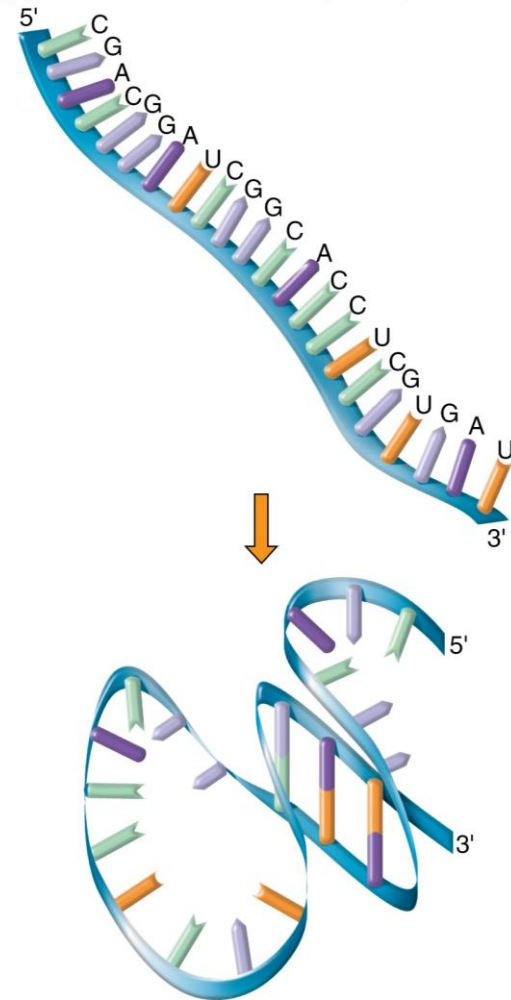
- Usually single-stranded



Complex folding pattern of RNA

Most RNAs are single stranded but can form base pairs within other parts of the same molecule

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6.4 DNA replication

Learning objectives:

- **Describe the key steps in the semiconservative replication of DNA**
- **Explain how the Meselson-Stahl experiment with “heavy” nitrogen showed that DNA replication is semiconservative.**
- **Summarize the key factors DNA polymerase requires to replicate DNA**
- **Outline the steps in the process of DNA replication and how they relate to the requirements of DNA polymerase.**
- **Discuss three ways cells preserve the accuracy and integrity of the genetic information in DNA.**

The model of DNA replication postulated by Watson and Crick

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Unwinding of double helix exposes bases on each strand

Each strand can act as a template for synthesis of new strands

New strand forms by insertion of complementary base pair

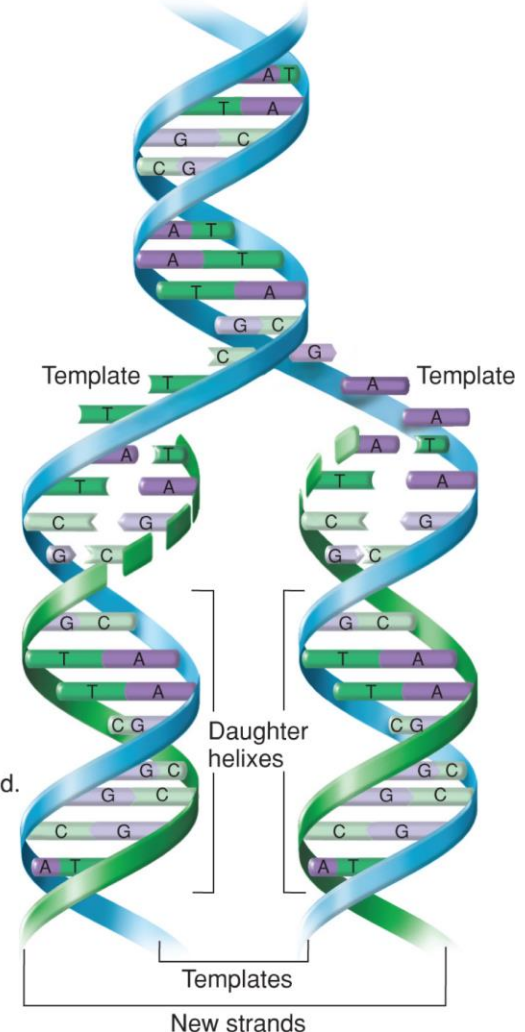
Single double helix becomes two identical daughter double helices

1. Original double helix

2. Strands separate.

3. Complementary bases align opposite templates.

4. Enzymes link sugar-phosphate backbone of aligned nucleotides into a continuous new strand.

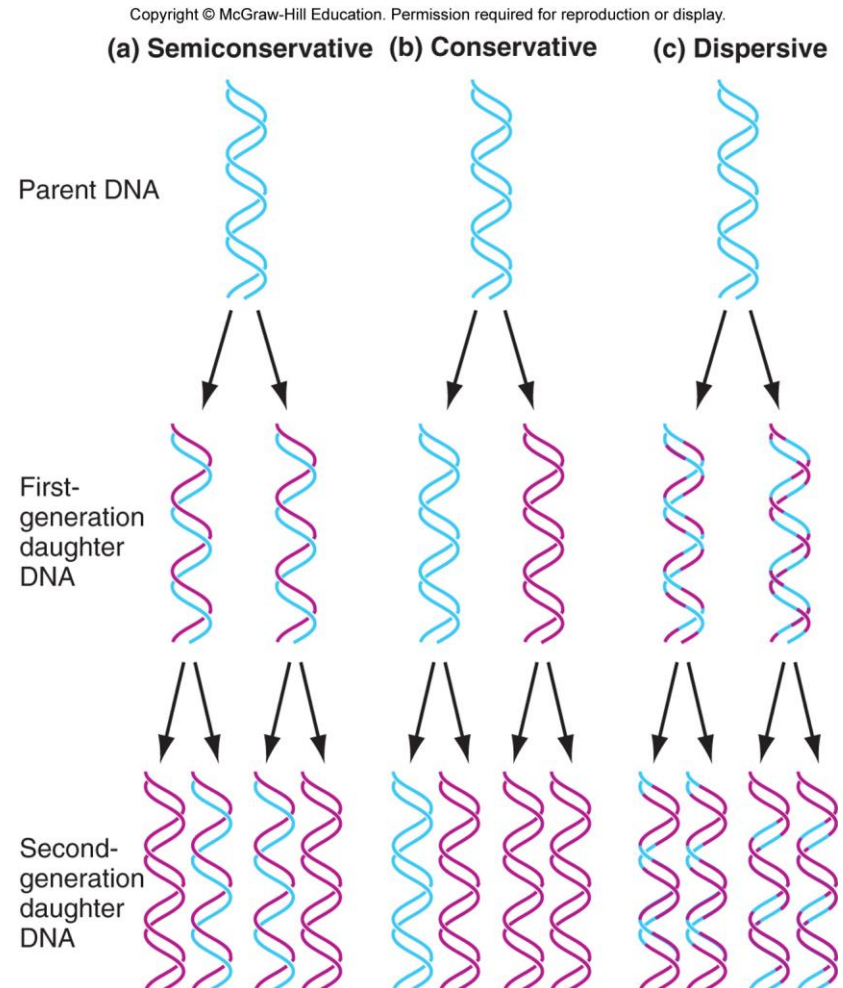


Three possible models of DNA replication

Semiconservative – the Watson-Crick model

Conservative – parental double helix remains intact, both strands of daughter helices are newly synthesized

Dispersive – both strands of both daughter helices contain original and newly synthesized DNA



The experimental approach to test the models of replication

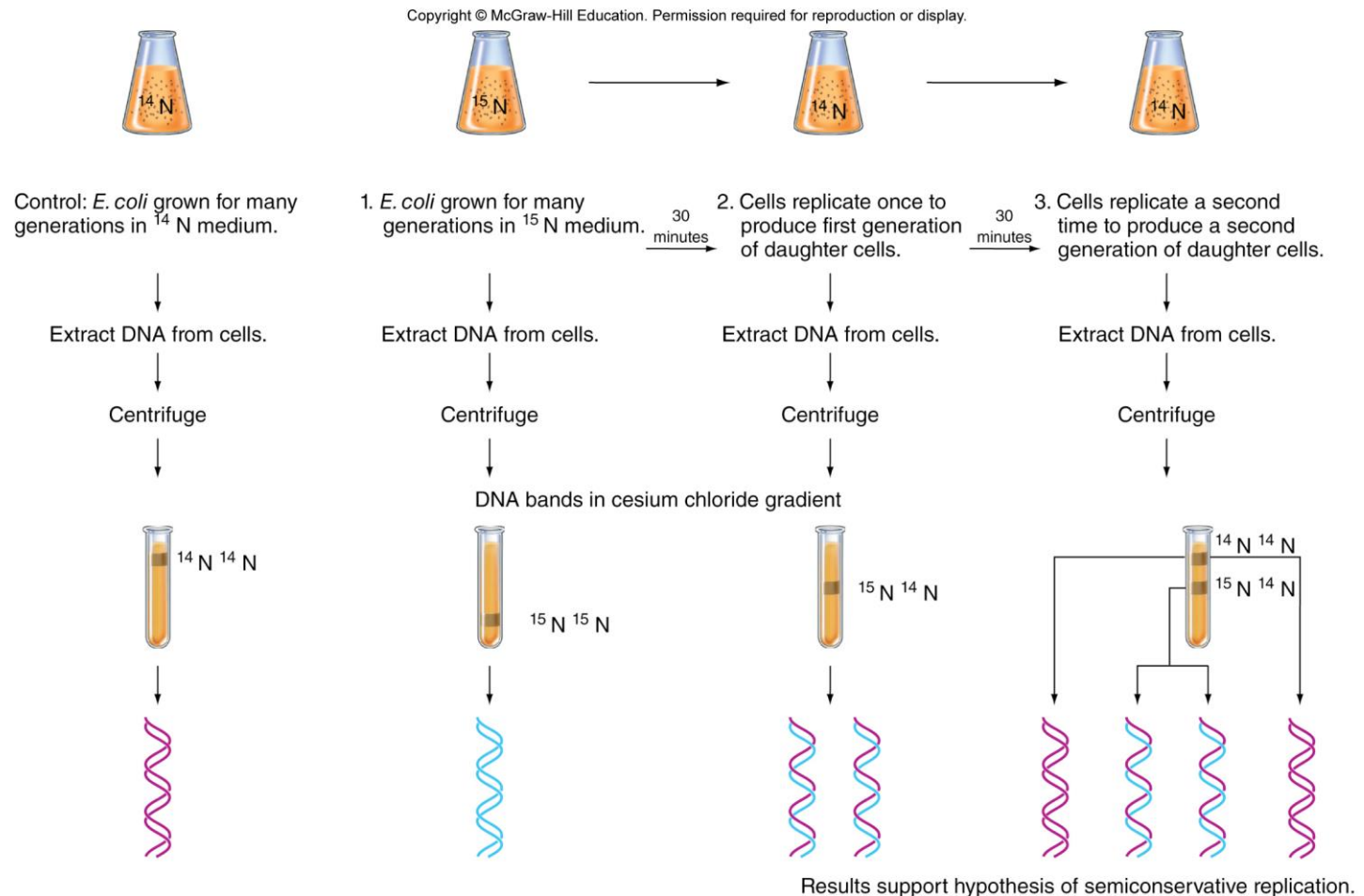
M. Meselson and F. Stahl (1958) separated preexisting "parental" DNA from newly synthesized daughter DNA

- **Grew *E. coli* in media containing ^{15}N (heavy isotope) then switched to media containing ^{14}N (normal isotope)**

Purified DNA from cells and subjected it to equilibrium density gradient ultracentrifugation

- **Cesium chloride (CsCl) forms stable gradient with highest density at bottom of tube**
- **DNA forms a tight band at position where its density equals the CsCl density**

How the Meselson-Stahl experiment confirmed semiconservative replication



The mechanism of DNA replication

A. Kornberg and others worked out the biochemical aspects of replication in *E. coli*

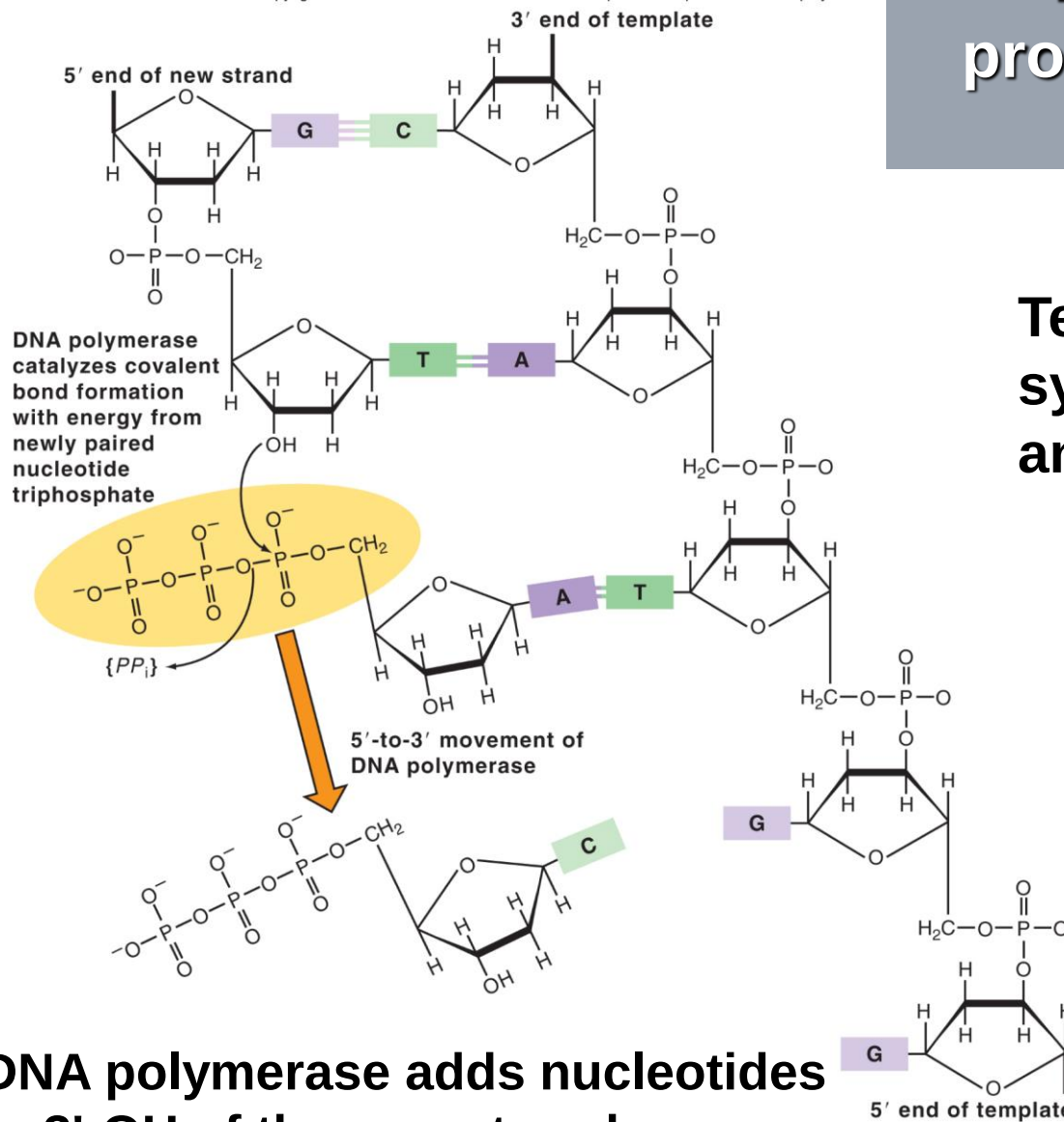
Energy for DNA synthesis comes from high-energy phosphate bonds associated with dNTPs

DNA polymerase (pol) catalyzes new phosphodiester bonds

Highly coordinated process has two stages

- **Initiation – proteins open up the double helix and prepare it for complementary base pairing**
- **Elongation – proteins connects the correct sequence of nucleotides on newly formed DNA stands**

DNA synthesis proceeds in a 5' to 3' direction



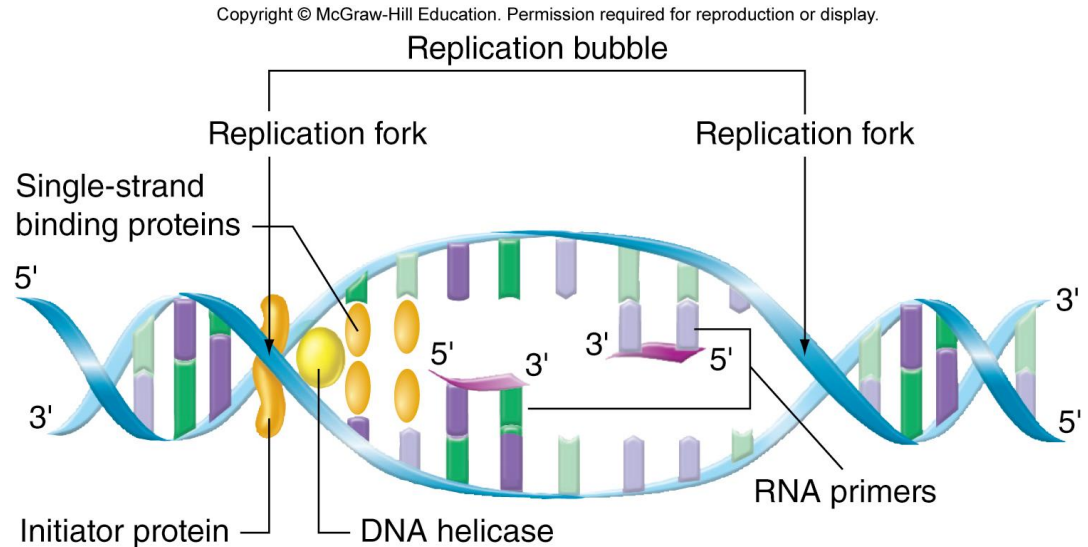
DNA polymerase adds nucleotides to 3'-OH of the new strand

The mechanism of DNA replication: Initiation

Initiator protein binds
to the origin of
replication

Helicase unwinds the
helix

**Single-strand binding
proteins** keep the
DNA helix open

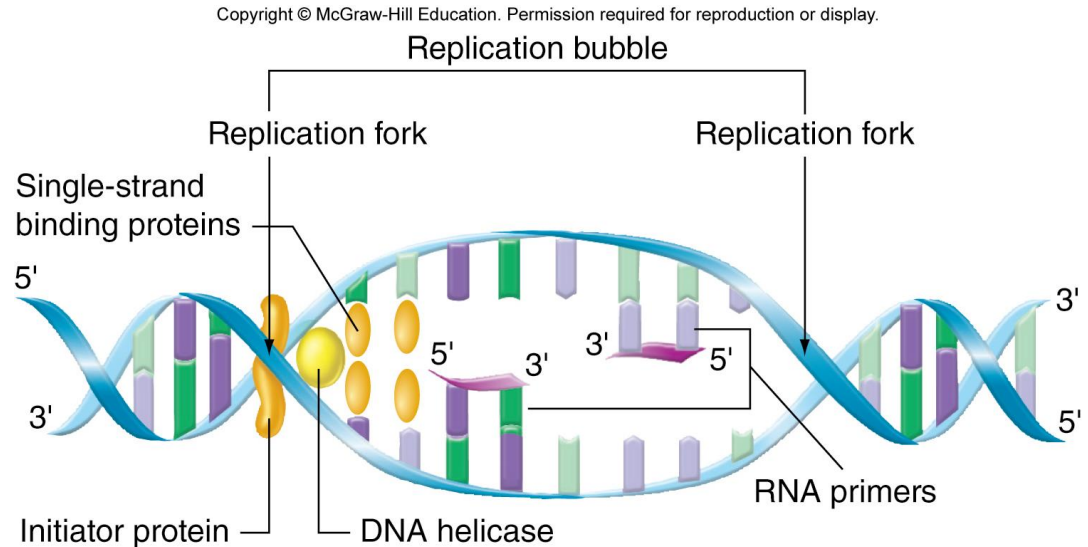


The mechanism of DNA replication: Initiation

Primase synthesizes
RNA primers

Primase synthesizes
RNA primer

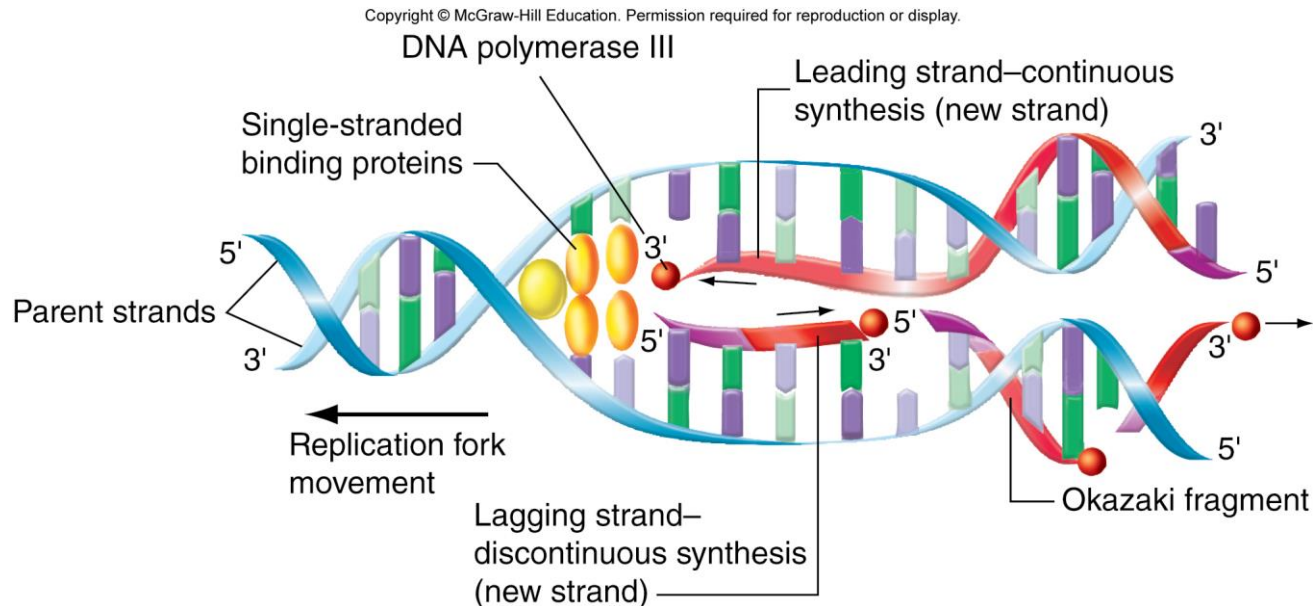
Primers are
complementary and
antiparallel to each
template strand



The mechanism of DNA replication: Elongation

The correct nucleotide sequence is copied from template strand to newly synthesized strand of DNA

DNA polymerase III catalyzes phosphodiester bond formation between adjacent nucleotides (polymerization)

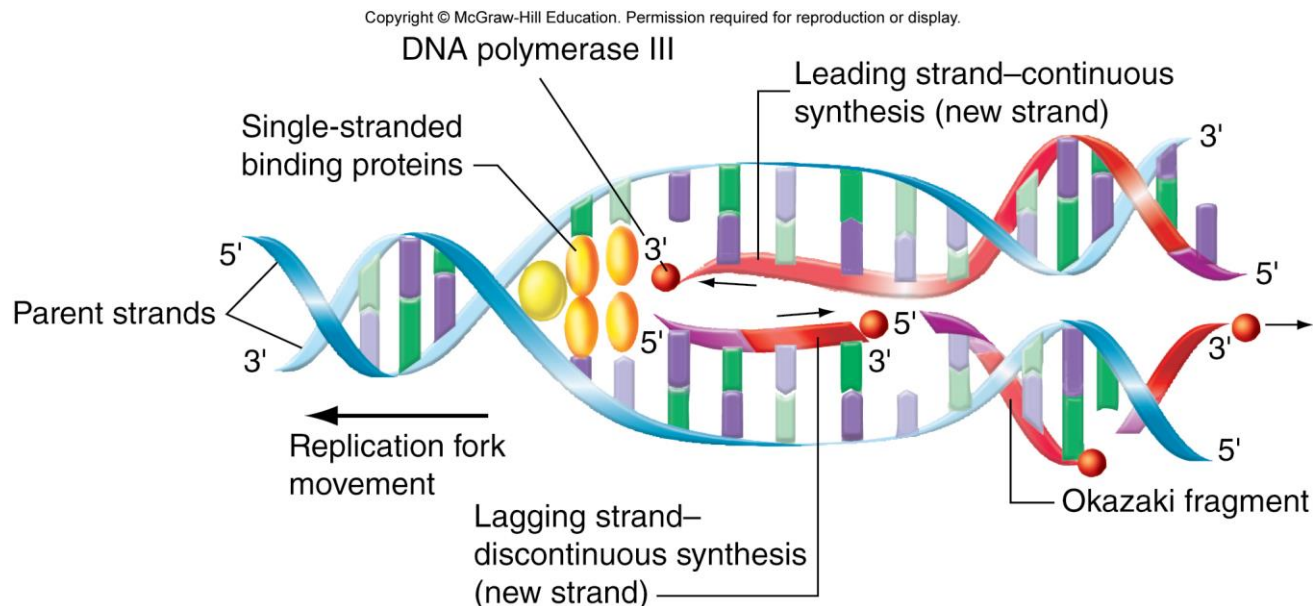


The mechanism of DNA replication: Elongation (cont)

Leading strand has continuous synthesis

Lagging strand has discontinuous synthesis

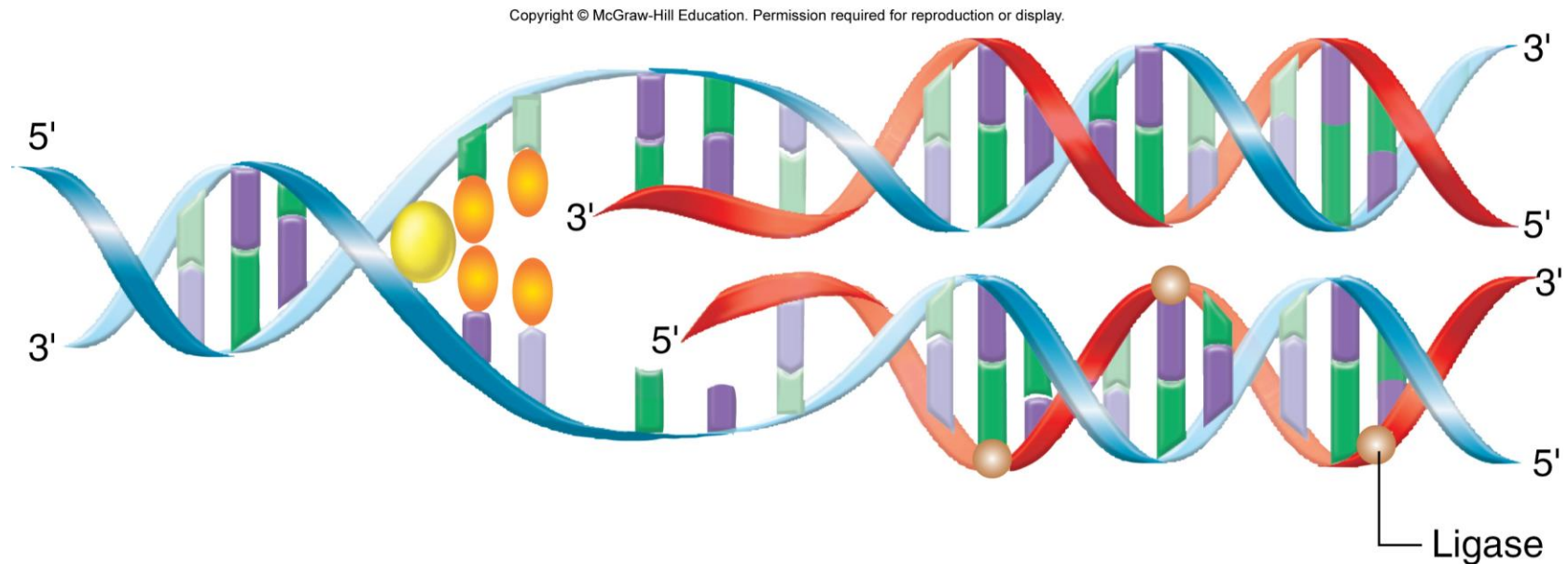
Okazaki fragment – short DNA fragments on lagging strand



The mechanism of DNA replication: Elongation (cont)

DNA polymerase I replaces RNA primer with DNA sequence

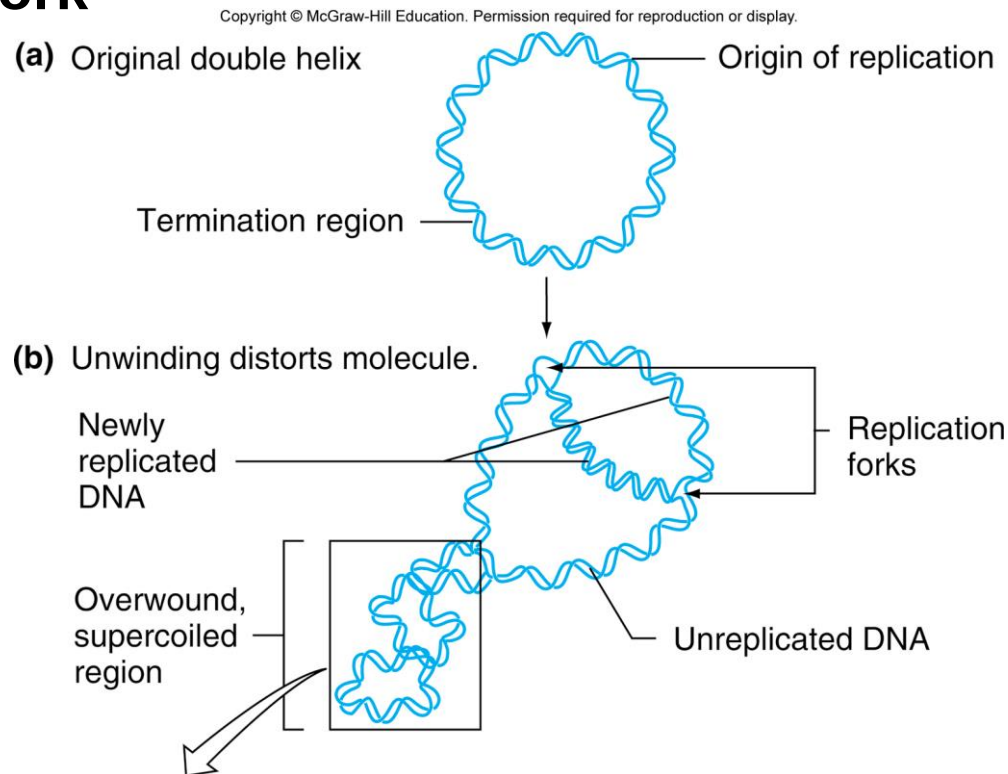
DNA ligase covalently joins successive Okazaki fragments together



The bidirectional replication of a circular bacterial chromosome: An overview

Replication proceeds in two directions from a single Ori

Unwinding of DNA creates **supercoiled DNA** ahead of replication fork



The bidirectional replication of a circular bacterial chromosome: An overview (cont)

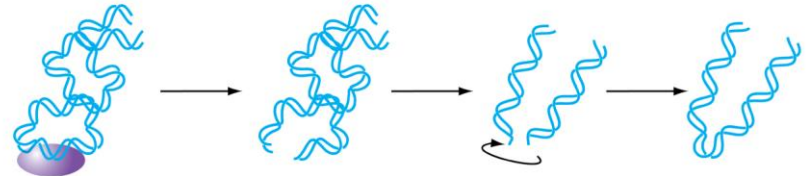
DNA topoisomerase relaxes supercoils by cutting the sugar phosphate backbone bonds strands of DNA

Unwound broken strands then sealed by ligase

Synthesis continues bidirectionally until replication forks meet

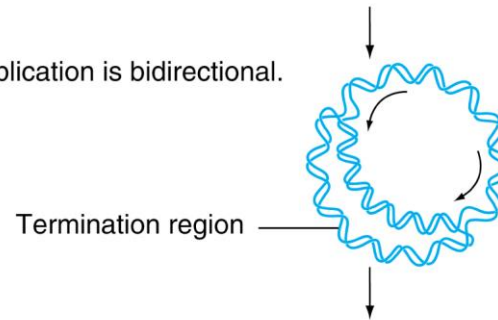
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(c) Topoisomerase relaxes supercoils by breaking, unwinding, and suturing the DNA.

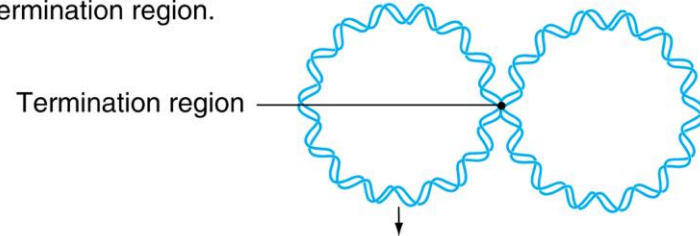


1. Topoisomerase in position to cut DNA
2. DNA cut by topoisomerase
3. Cut strands rotate to unwind
4. Cut ends of strands rejoined by ligase

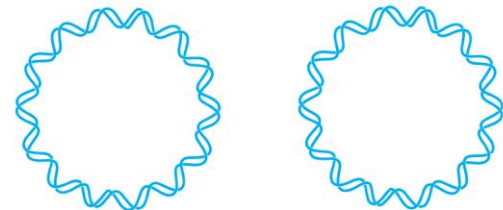
(d) Replication is bidirectional.



(e) Replication is complete when replication forks meet at the termination region.



(f) Topoisomerases separate entwined daughter chromosomes, yielding two daughter molecules.



Cells must ensure accuracy of genetic information

Three ways to ensure fidelity of DNA information:

Redundancy – either strand of the double helix can specify the sequence of the other strand

Precision of cellular replication machinery

- **DNA polymerase I and III have proofreading ability (more about this in Chapter 7)**

DNA repair enzymes (described in Chapter 7)

6.5 Recombination at the DNA level

Learning objectives:

- Summarize the evidence from tetrad analysis confirming that recombination occurs at the four-strand stage and involves reciprocal exchange.
- Explain how we know that DNA breaks and rejoins during recombination.
- List the key steps of recombination at the molecular level.
- Explain why recombination events do not always result in crossing-over.
- Describe how mismatch repair of heteroduplex regions can lead to gene conversion in fungal tetrads.

Recombination at the DNA level

New combinations of alleles are created by two types of events in meiosis

Independent assortment – each pair of homologous chromosomes segregates freely from the other (Chapter 4)

- **Creates new allele combinations for unlinked genes**

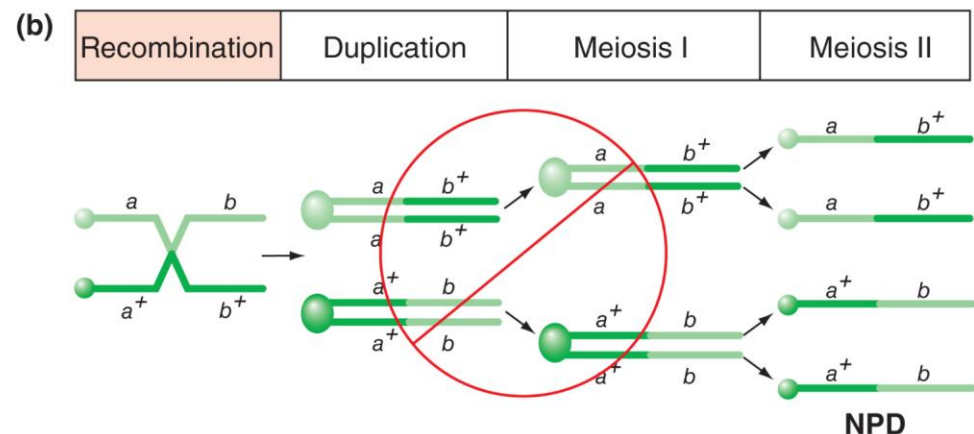
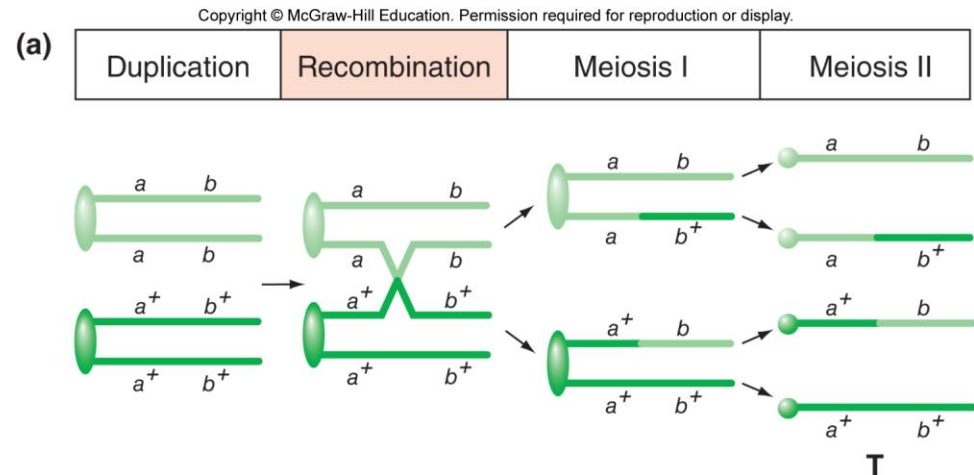
Crossing over – two homologous chromosomes exchange portions of DNA (Chapter 5)

- **Creates new allele combinations for linked genes**
- **Ensures proper chromosome segregation during meiosis**

Tetrad analysis indicates recombination occurs at the four-strand stage by reciprocal exchange

T tetrads result from recombination at the four-cell stage.

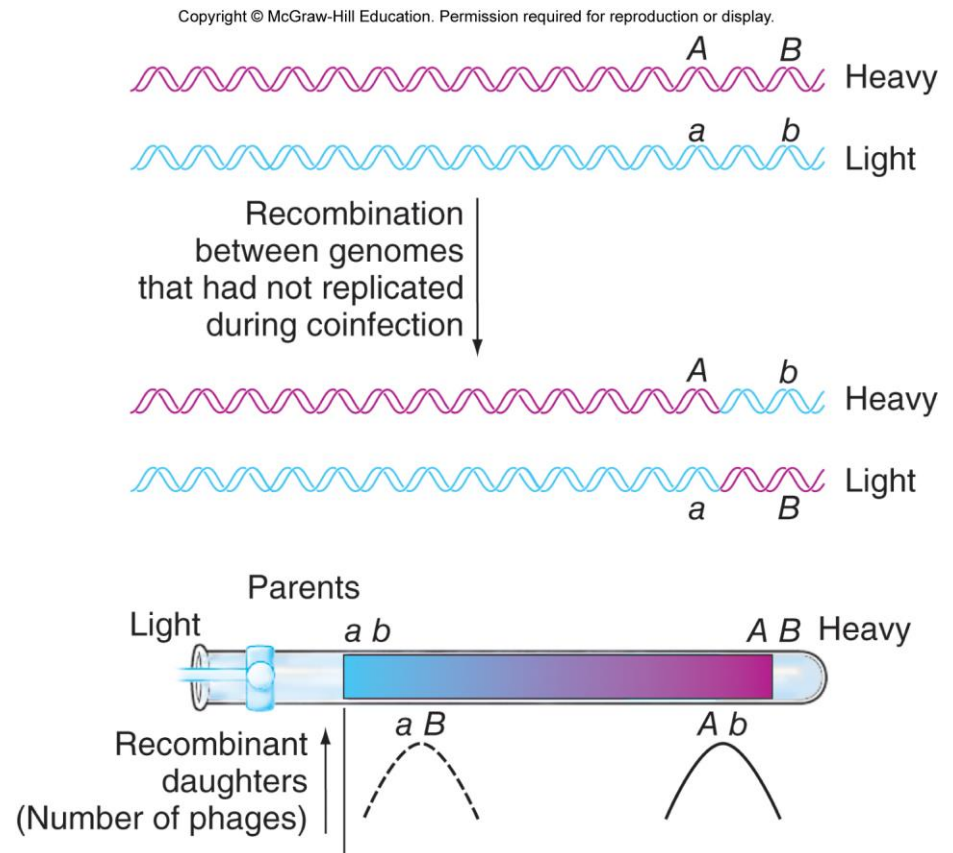
T and NPD tetrads contain an equal number of both classes of recombinants. See Fig. 5.20



DNA molecules break and rejoin during recombination: The experimental evidence

Meselson and Weigle coinfecting *E. coli* with A B and a b phage.

- Phage A B grown with heavy isotopes (^{13}C and ^{15}N)
- Phage a b grown with light isotopes (^{12}C and ^{14}N)
- If DNA breaks and rejoins, expect to see some A b phage with mostly heavy DNA



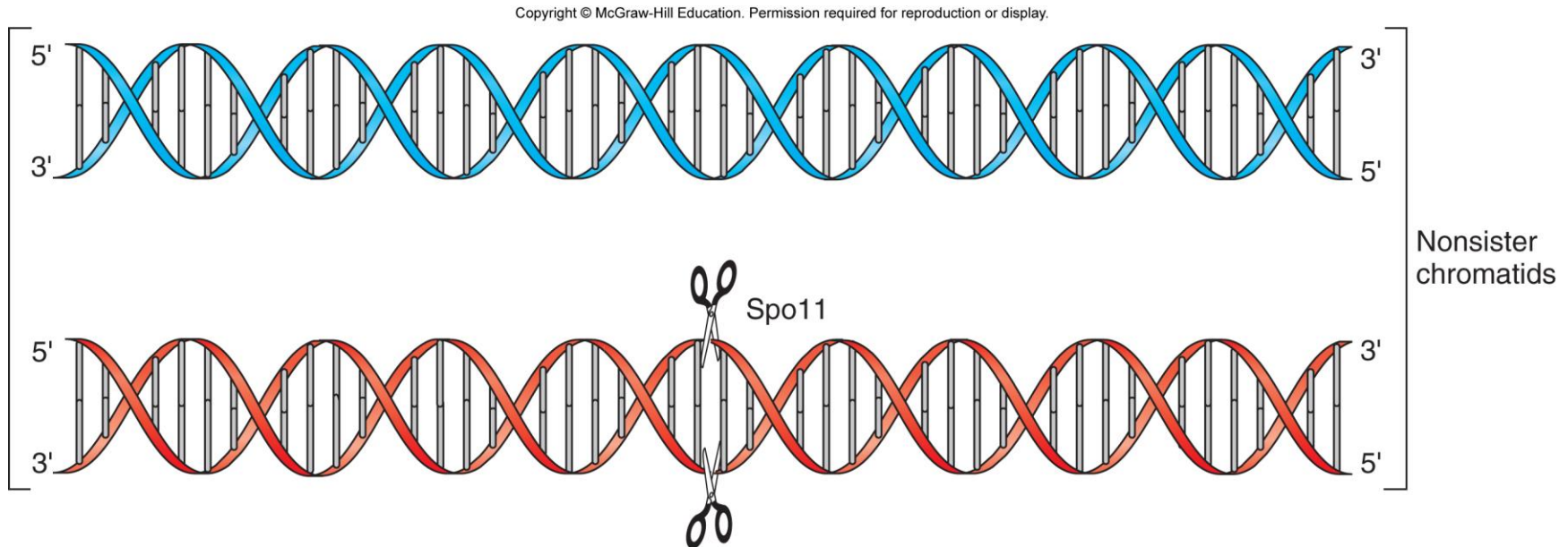
Recombination at the molecular level

Recombination is initiated by a protein that is conserved in many organisms, Spo11.

It is likely that the mechanism of recombination is also conserved in all eukaryotes.

Step 1 in the model of recombination: Double-strand break formation

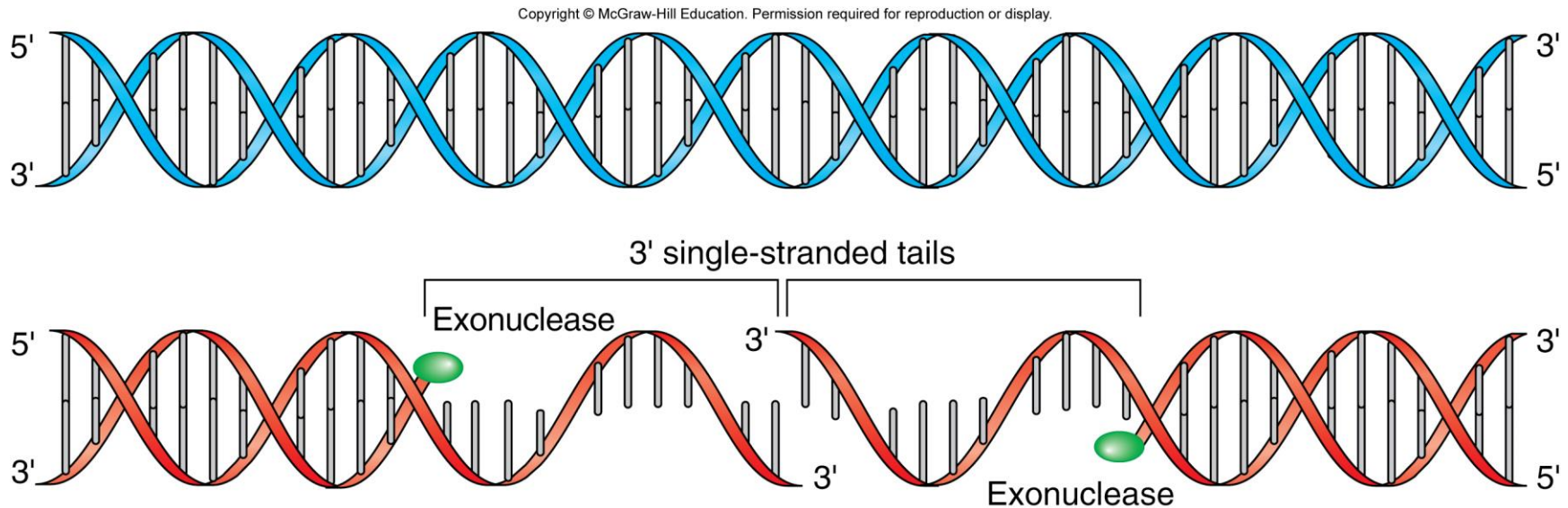
Spo11 breaks phosphodiester bonds of both strands of one chromatid



Two non-sister chromatids are shown.

Step 2 in the model of recombination: Resection

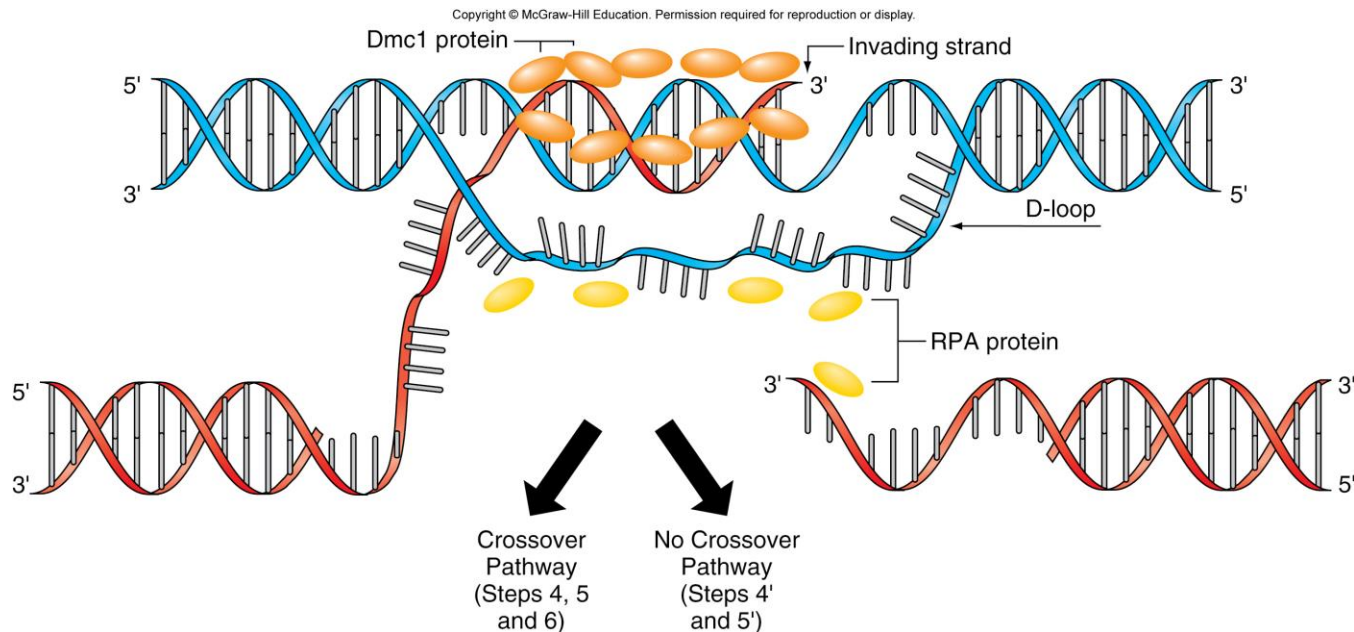
5' ends of each broken strand are degraded by an **exonuclease** to create 3' single-stranded tails.



Step 3 in the model of recombination: strand invasion

Dmc1 helps one single-strand tail invades a non-sister chromatid and forms stable **heteroduplex**

Displacement loop (D-loop) from invaded chromatid is stabilized by single-strand binding protein

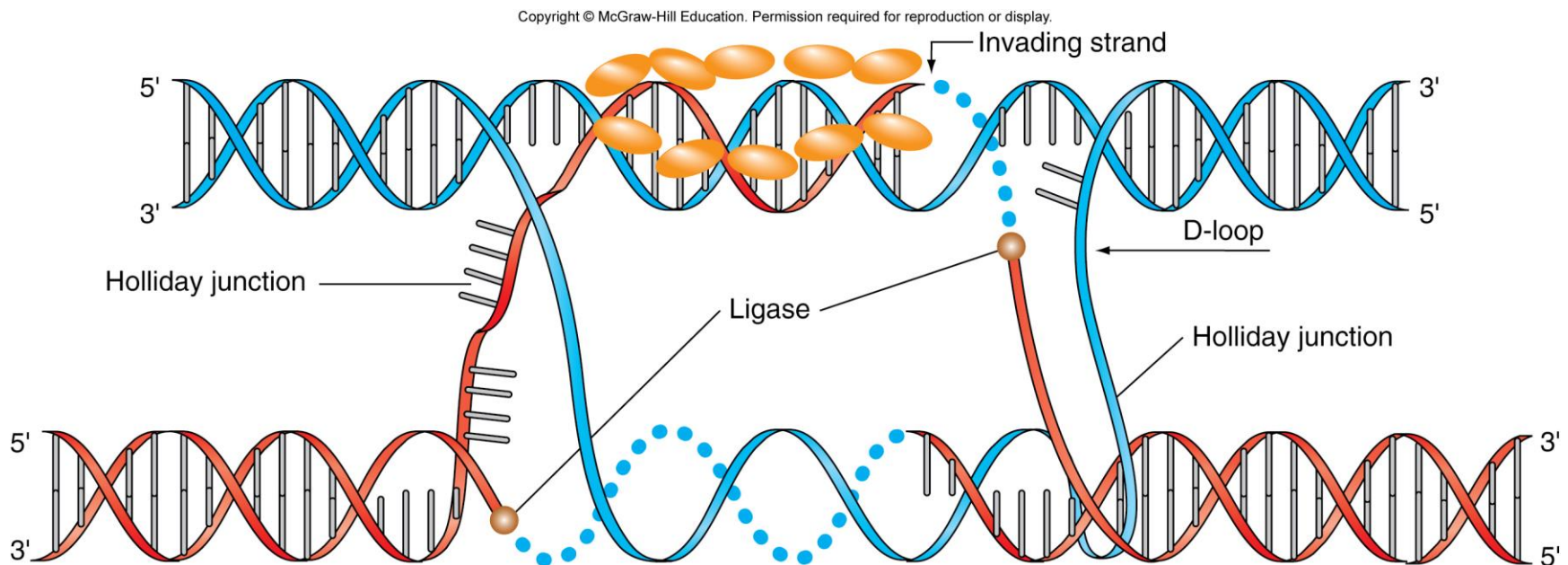


Step 4 in the model of recombination: Formation of double Holliday junction

D-loop and other 3' tail form a second heteroduplex.

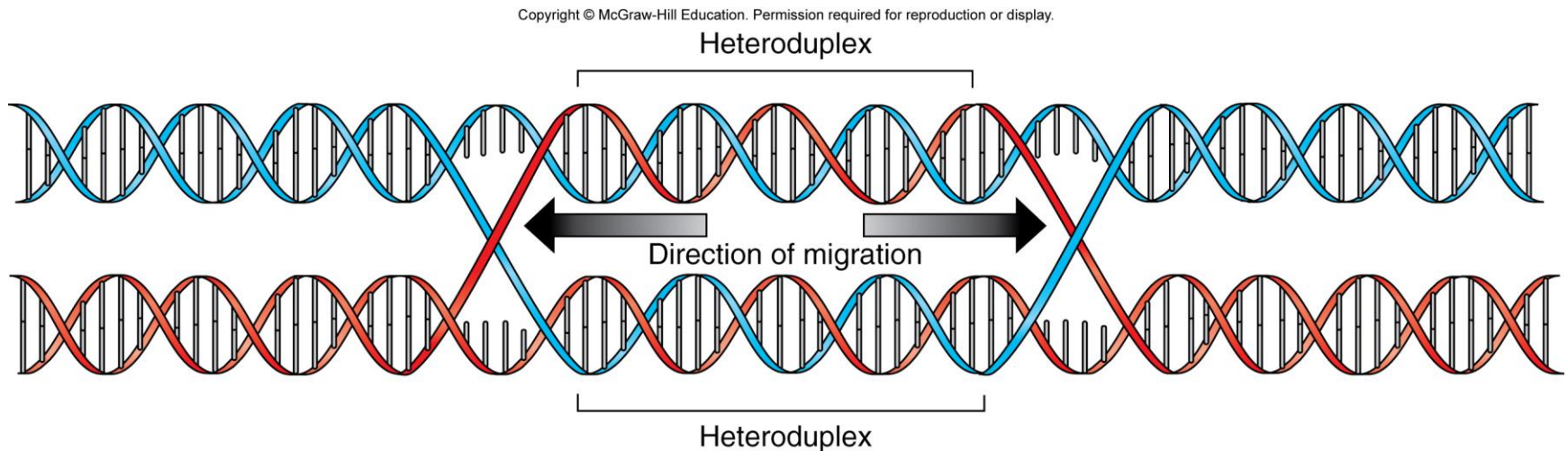
DNA synthesis replaces the DNA degraded during formation of the 3' tails.

DNA ligase reseals the DNA backbones.



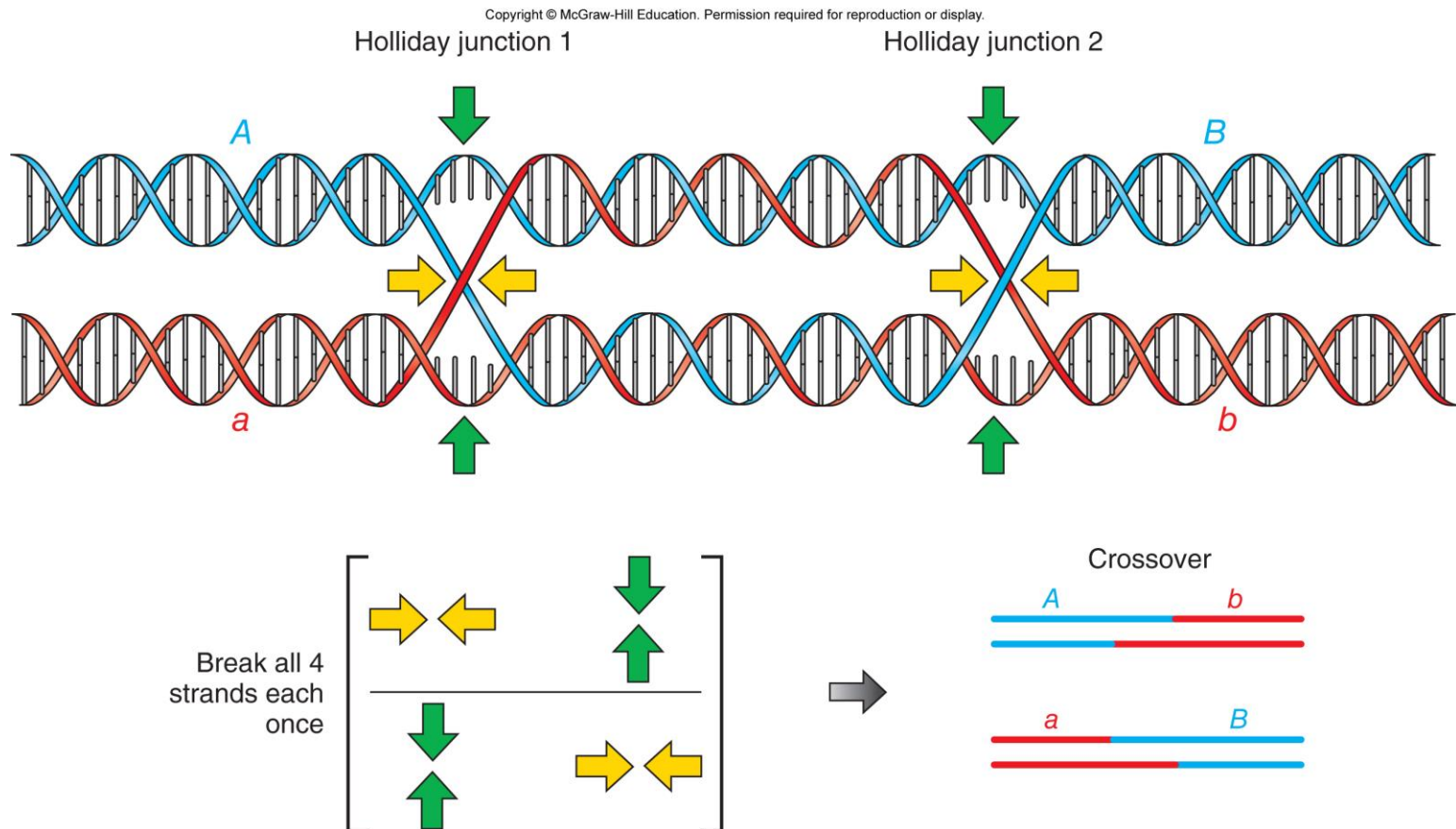
Step 5 in the model of recombination: Branch migration

Heteroduplex region of both DNA molecules is lengthened



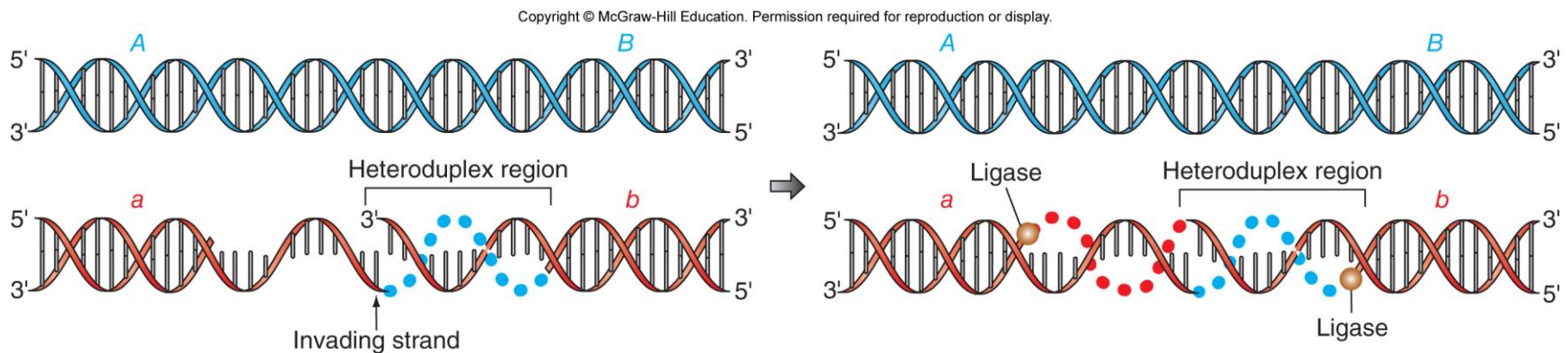
Step 6 in the model of recombination: Resolution of the Holliday junction

Holliday junctions are resolved by resolvase and ligase .



Recombination without crossing over

Anticrossover helicase disentangles the invading strand from the nonsister chromatid preventing Holliday junction formation.



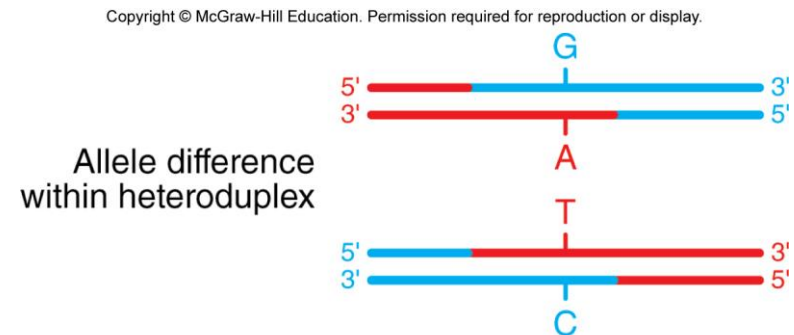
Heteroduplex regions occur at sites of genetic exchange

Two strands of DNA don't break and rejoin at the same location

- Breakpoints on each strand can be 100s-1000s bp apart

Heteroduplex – region of DNA between breakpoints

- One strand is maternal and other is paternal
- Strands can have mismatches



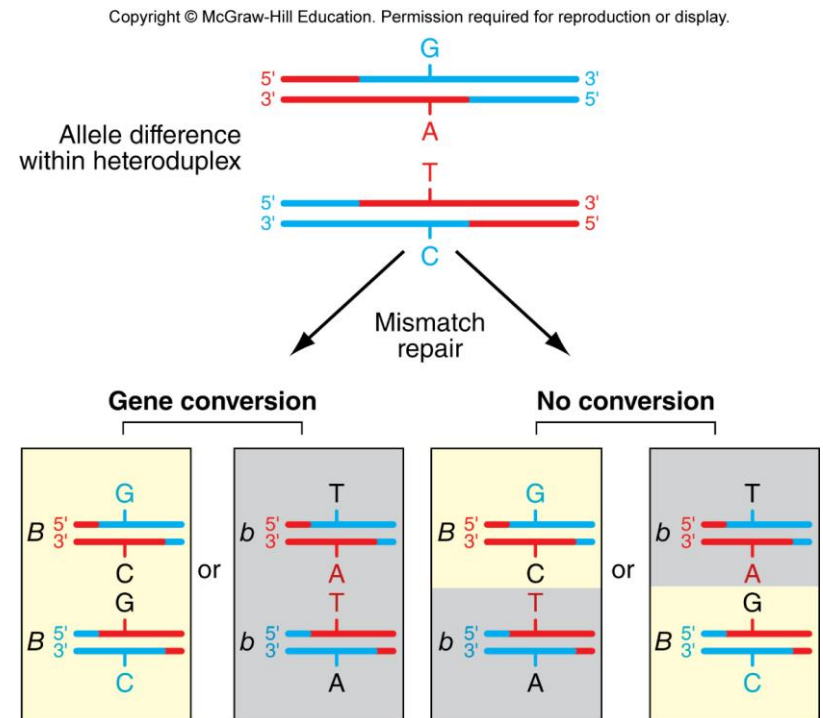
Mismatches in heteroduplexes can be repaired

DNA repair enzymes eliminate mismatches

Either allele can be converted

Gene conversion – deviations from expected 2:2 segregation, e.g. 3:1 or 1:3

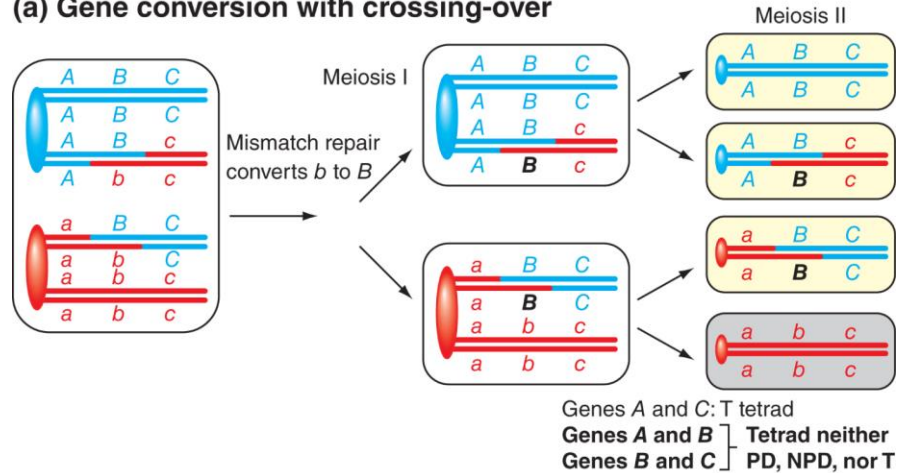
In yeast, gene conversion occurs 50:50 with and without crossing over of flanking markers



Detection of gene conversion in yeast tetrads

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(a) Gene conversion with crossing-over



(b) Gene conversion without crossing-over

